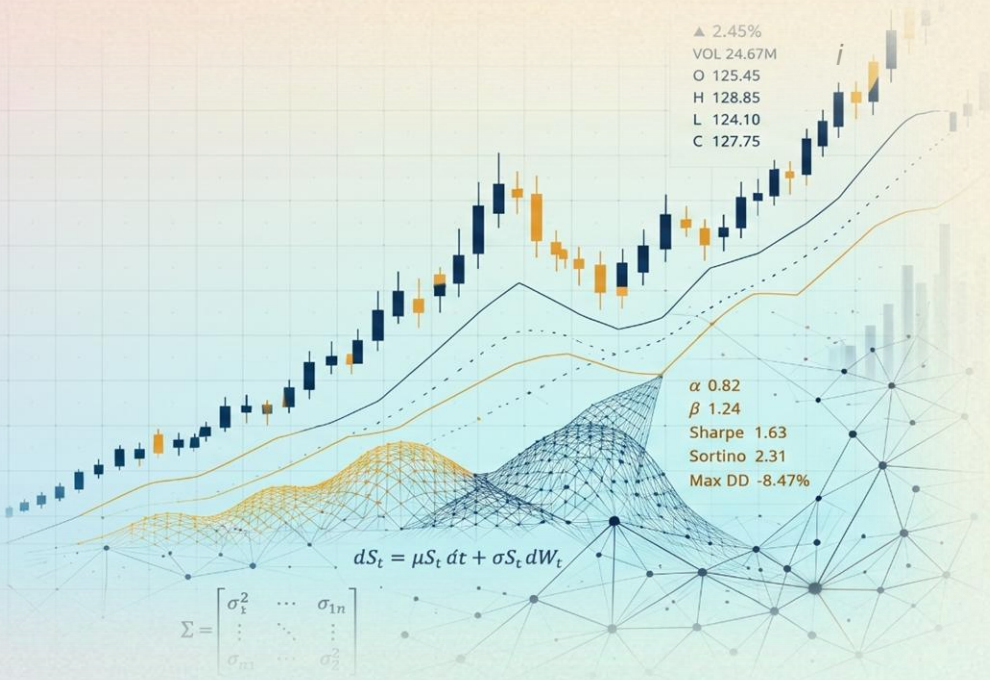
WORLDQUANT UNIVERSITY



A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT

Dynamic Expected-Loss Optimization in Credit Risk

An Explainable & Fair Machine-Learning Framework

~Part II~

BY

Sana Ur Rehman Arain

DEPARTMENT

Master of Science in Financial Engineering

SUPERVISOR

Prof. Ivan Blanco

2026



"In God we trust, all others must bring data."

W. Edwards Deming.

PREFACE

Beyond the 10,000-record challenge

Part One of this research began with a deliberately constrained question: could meaningful credit-risk modelling be carried out using a dataset of only 10,000 loans, severe class imbalance, and a limited number of resolved default events? Rather than treating those weaknesses simply as reasons to abandon the exercise, I chose to work through them. The objective was not to construct a flawless banking system, but to understand exactly where a machine-learning credit model begins to fail, why those failures occur, and which methodological choices become necessary when the data are imperfect.

That experiment produced a qualified answer. The small dataset was sufficient to demonstrate that machine-learning models could extract useful risk ordering and that SHAP could provide meaningful technical explanations of their predictions. It also revealed something more important: successful classification is not the same as reliable financial risk estimation. Synthetic oversampling could improve minority-class detection while distorting the probability distribution. A model could rank borrowers reasonably well while its probability estimates remained unsuitable for monetary Expected Loss calculations. A random train-test split could measure performance on unseen observations without answering the more difficult question of whether the model would remain reliable on borrowers originating in a later period. A statistically convenient threshold could produce classifications without representing an economically justified lending rule.

Those limitations became the roadmap for Part Two.

The central question therefore changed. Part Two is no longer primarily concerned with whether a machine-learning model can identify risk. It asks what happens after that model exists. Can its probabilities be trusted? Do they remain aligned with observed outcomes over time? How should a probability estimate be translated into a financial decision? What happens when the borrower population or observed risk level changes? Can the resulting decisions be explained and audited? And what trade-offs become visible when predictive, financial and proxy-group outcomes are examined together?

To investigate these questions, the original 10,000-record snapshot was replaced with a substantially larger historical Lending Club cohort containing 603,589 resolved 36-month loans issued between 2011 and 2015. The change in scale made it possible to abandon the random-split structure of Part One and impose a chronological design: loans from 2011–2013 are used for model development, 2014 for validation and probability recalibration, and 2015 as the final out-of-time evaluation cohort. The purpose of this structure is not to claim that the study reproduces a live lending system, but to create a more demanding historical test of how a fixed modelling framework behaves when it is carried forward to a later loan vintage.

Part Two also changes the role of the model itself. Logistic Regression remains as an interpretable benchmark, while LightGBM is evaluated as the non-linear challenger. However, model selection is no longer the end of the analysis. Discrimination is followed by probability calibration, population-stability diagnostics, calibration-drift analysis and sensitivity testing. Calibrated charge-off probabilities are then combined with a retrospective historical loss-severity assumption and funded amount as an exposure proxy to construct an Expected Loss decision rule. This allows the research to examine not simply which loans receive higher risk scores, but how a frozen risk policy changes the composition, realized principal loss and undiscounted cash return of a historical portfolio.

The governance questions raised in Part One are also carried forward more cautiously. TreeSHAP is used to describe both global and local behaviour of the underlying LightGBM model, while home ownership is examined only as an illustrative socioeconomic proxy in group-level sensitivity and error-rate analysis. These exercises are intended to demonstrate model-governance methods, not to claim regulatory compliance, causal explanations, or a definitive protected-class fairness assessment.

In this sense, Part Two is not a replacement for Part One. It is the consequence of it. The first study was valuable precisely because its constraints exposed the difference between obtaining a prediction and building a defensible financial decision framework. The present study takes those limitations seriously and asks how far the original modelling idea can be extended when the dataset is larger, outcomes are mature, time is treated explicitly, financial consequences are incorporated, and the model is subjected to a broader set of validation and governance tests.

The progression from Part One to Part Two therefore reflects a change in emphasis: from **classification to calibration, from random evaluation to chronological validation, from statistical thresholds to Expected Loss decisions, and from model explanation to model governance.**

The aim remains deliberately modest. This is not a validated production lending system, nor is it presented as one. It is a reproducible historical research framework designed to examine what must be considered before a predictive credit model can responsibly be connected to financial decisions. If Part One asked, “Can meaningful modelling be done under severe constraints?”, Part Two asks the question that naturally follows:

“Once a model can rank risk, what is required before its predictions can be trusted for financial decision-making?”.



ABSTRACT

To rank the borrowers and applicants based on expected risks, accurate credit-risk estimation should preserve adequate calibration over time and be tied to economic decision criteria. Continuing previous research into comparing traditional and explainable machine learning models, this study proposes a replicable, cost sensitive credit-risk framework, based on the sample of 603,589 completed three-year loans from Lending Club provided in 2011–2015. The chronological design is adopted where 2011-2013 data is used for model training, 2014 year serves as validation and probability recalibration period, while 2015 is set aside for out-of-time testing.

LightGBM challenger (unweighted) managed to attain the ROC-AUC of 0.6625 for 2015 OOT test, which is slightly higher than 0.6569 for logistic regression. LightGBM probabilities were recalibrated via isotonic regression on the 2014 validation set. These probabilities, along with exposure-weighted historical loss-given default assumption of 50.05%, and funded amount as simplified exposure measure were used to build an Expected Loss decision rule. With the 5.00% estimated principal loss target, the solver selected calibrated probability of default of 0.1696 as the loan approval threshold. When applied unchanged to the 2015 year, this policy kept 68.92% of observed originated loans with 5.27% principal loss rate, compared to 7.36% for accept-all benchmark. This loan portfolio yielded 7.18% net cash return rate vs. 6.35% for accept-all.

Results from temporal diagnostics showed that discrimination could exist in a relatively stable state while there was a simultaneous deterioration in probability calibration. It was found from the Kitagawa decomposition that the rise in the charge-off rates was mainly due to deterioration in segments and was not associated with changes in the portfolio mix, but without identifying any causal reason for it. TreeSHAP provided global and local technical insights into the underlying LightGBM model, and the proxy group fairness audit of the frozen decision rule gave a renter/ non-renter False Positive Rate ratio of 1.8895. Feature-removal sensitivity analysis showed that proxy-group outcomes could depend significantly on the chosen feature policies, but renter is still a socioeconomic proxy, and the analysis is not a protected class fair-lending evaluation.

In summary, it is seen that discrimination, calibration, temporal stability, financial consequences, explainability, and proxy-group fairness should all be considered together when analyzing machine learning-based credit risk models. The presented system must be viewed as a transparent historical policy simulation and model governance research tool, but not as a valid lending system.

Keywords: Credit Risk Modelling; Machine Learning; Expected Loss; LightGBM; Probability Calibration; Out-of-Time Validation; TreeSHAP; Model Explainability; Population Drift; Proxy-Group Fairness; Financial Decision Analysis.

Note on Reproducibility: For full academic transparency and auditability, the entire system was implemented with open-source Python libraries such as `scikit-learn` for baselines modeling and pipeline preprocessing, `LightGBM` for gradient boosting and `SHAP` for game theoretic explainability. Deterministic computation was ensured through setting up the fixed random seed (`random_state=42`) across all sampling, chronological splitting, and modeling algorithms. The whole source code base, data snapshot at different points in time, and frozen JSON policy rules are available in the project repository *Arain, S. U. R. (2026). Dynamic Expected-Loss Optimization in Credit Risk: An Explainable and Fair Machine-Learning Framework – Part II (Version 1.0)* [*Software*]. *https://github.com/sanaurrehmanarain/dynamic-credit-risk-xai*.

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CHAPTER 1: INTRODUCTION

1.1. Background

The estimation of credit risk in a precise manner constitutes the core of modern banking practices. The progress in machine learning technology and its adoption to tabular financial data made credit scoring move from classical scorecards to sophisticated non-linear models. However, the majority of standard machine learning approaches consider default as a static binary classification problem. Although this methodology successfully ranks borrowers according to their risks, it does not necessarily reflect the dynamic, dollar-value aspect of portfolios' management. The current report represents the second methodological step of the work after identifying the main issues related to the topic in the previous study. Specifically, it follows Arain, S. U. R. (2026, October 14). *Machine learning for credit risk modeling: A comparative study of traditional & explainable AI approaches - Part I.* SSRN. https://doi.org/10.2139/ssrn.7614558. Unlike the first study that was limited only to a 10,000-record dataset snapshot, this framework considers a developed sample of 603,589 36-month loans issued by Lending Club between 2011 and 2015.

1.2. Motivation and Problem Statement

There are two major limitations of conventional ML approaches for credit risk modelling. Firstly, they use naive and uncalibrated decision threshold (usually 0.50) and probabilities that do not reflect the real financial risk. Secondly, credit population is a dynamic set since macroeconomic environment and underwriting criteria change over time. Exploratory analysis of the Lending Club sample shows that the charge-off rate increased from 12.52% in 2011-2013 to 14.89% in 2015.

One persistent flaw in academic credit risk models is the practice of random train/test splits and static snapshots of the data, which may lead to overly optimistic forecasts of the future and not account for distribution shifts over loan cohorts. Based on the initial proof of concept in Arain (2026), this study attempts to address some of those issues by scaling up to a much larger set of loans that have been completely resolved, imposing chronological out-of-time validation, splitting model features into pre-origination and post-origination groups, avoiding synthetic resampling for the final probability model, calibrating probabilities, using charge-offs based on calibrated probabilities under an expected loss decision rule, and finally using TreeSHAP and proxy group sensitivity analysis for model governance. This framework is thus not to be taken as a validated blueprint for model implementation but a reproducible environment to simulate and validate decisions.

In order to solve the outlined problems, the main methodological task of the present project could be stated in the following way: *How can a credit-risk modelling framework estimate the probability of a recorded final charge-off outcome using leakage-free features and chronological validation, considering discrimination, calibration, population drift, explainability, fairness sensitivity, and financial decision making aspects?*

1.3. Research Questions

In order to answer the main research question, the following research questions will be explored:

1. How do Logistic Regression and LightGBM models compare in terms of discrimination on validation and out-of-time test datasets?
2. How does the model performance vary depending on chronological evaluation across loan vintages to take population drift into account?
3. How do borrower and application level features compare in their predictivity to lender-provided grade and pricing features?
4. Are the estimated probabilities well-calibrated for risk estimation, and is the use of isotonic regression beneficial for out-of-time results?
5. How do different feature policy decisions (such as excluding socioeconomic features) affect the discrimination rates?
6. Which features are the most important for LightGBM model predictions, and how can the explanation for them be understood?
7. How is the risk estimate, which is obtained through calibration, related to a concept of Expected Loss and approval decisions?
8. What restrictions exist for using originated and resolved Lending Club loans and their proxy variables?

1.4. Research Objectives and Contributions

The above framework brings in a reproducible research framework which attempts to replicate certain production constraints.

- **Out-of-Time (OOT) Validation:** To analyze the temporal robustness when varying the vintages of loans and characteristics of the portfolio by a chronological split consisting of Training (2011-2013), Validation (2014), and Held-Out Test (2015). The chronological design takes into account the impact of time distribution shift but does not attribute it to any particular macroeconomic factor.



- **Probability Calibration:** To test how well the raw probabilities of the models match the empirical rates of charge-offs as well as the impact of the probability recalibration estimated solely from 2014 validation data and then used for 2015 out-of-time test.
- **Cost-Sensitive Optimization:** To aggregate calibrated charge-off probabilities, historical losses, and the approximation of exposures through the funded amount in order to create an Expected Loss based decision-making rule with constraints on portfolio risk.
- **Explainability & Fairness:** The decision rule is also analyzed in terms of proxy group-fairness using some socioeconomic indicators such as home ownership status. The analysis is not claimed as an assessment of disparate-impact since there is no protected-class information.

Methodologically, the key point of this framework is the transition from a credit scoring task which requires a ranking function to a dynamic financial decision solver. Using post-origination recovery and repayment data from historical loan files (`payment_outcomes_2011_2015_36m.parquet`), a retrofitted Loss Given Default (LGD) is estimated. Together with funded exposure, the Exposure at Default (EAD) it gives:

$$EL = PD \times LGD \times EAD \tag{1}$$

This equation allows to formulate a decision rule which maximizes approved loan volume subject to a portfolio risk appetite constraint.

1.5. Regulatory and Ethical Context

Use of machine learning in credit underwriting requires strict compliance with regulatory transparency requirements. While algorithms like LightGBM are capable of capturing complex non-linear borrower interactions, there is a danger of creating a black-box algorithm. Per the Equal Credit Opportunity Act (ECOA), lenders are obliged to provide specific Adverse Action notifications that give the precise reasons for credit denial. Optimization of financial decision making can lead to different decisions across proxy groups of the population even if the policy is imposed for all observations. In this case, sensitivity analysis suggests that choice of features and policy can influence actual outcomes in terms of principal losses and influence renter error rates relative to non-renters. These results motivate a governance analysis of predictive, financial, and proxy-group outcomes rather than establishing a causal fairness-profitability trade-off.

1.6. Scope and Delimitations

The study scope is limited exclusively to originated 36-month loans with known outcomes (Fully Paid or Charged Off). The modeling does not include rejected applications because repayment outcomes

are impossible to ascertain. At the same time, loans with unknown intermediate statuses (Current, Late, In Grace Period) are excluded intentionally. The model predicts the probability of the known final charge-off outcome in this particular Lending Club cohort. In this report, this metric is denoted as Probability of Default (PD) for simplicity. However, it is not a standard fixed-horizon regulatory PD and does not predict rejected applications. Furthermore, the target is not the probability of default over the 12-, 24-, or 36-month observation window and does not incorporate the time-to-default relationship. It should therefore only be understood as the probability of the final status of Charged Off in the chosen historical sample of originated and resolved loans.

Several methodological delimitations constrain the validity of claims above:

- **Feature Availability:** This framework uses historical data, making the assumption that all features for the input to the model were available at the time of the credit decision. The exact timeline of availability of specific feature, in particular `verification_status`, before the decision has not been verified. In this regard, the baseline model should only be seen as a retro research specification, conditional on the availability of these predictors. Should it later be proven that `verification_status` or some other field was not actually available at the time of the decision, the feature would have to be dropped from the model along with the entire process of calibration and decision-making.

- **Validation Constraints:** Chronological OOT split checks the robustness against shifting populations, while taking into account the constraint of delayed labeling: In real production environment the whole 36-month results of 2014 loans cannot be known until January 2015.

- **Financial Proxies:** The framework for the Expected Loss (EL) is essentially the retrospective simulation based on the information from the historical outcomes. The original loan amount is used as a simplification of the Exposure at Default (EAD) metric. Also, although net cash flows are calculated (taking into account recoveries and collection fees), the framework calculates undiscounted cash-based metrics, ignoring the internal costs of funding, servicing, and time-value of money.

- **Fairness and Explainability:** Algorithmic fairness in this case is performed on home ownership stratification of borrowers, which can be seen as a proxy rather than a verifiable socio-economic characteristic. Furthermore, the use of SHAP to explain technical details of the model behavior and features' correlations does not imply the existence of any economic causality and is not sufficient for the Adverse Action notice in compliance regulations.



1.7. Project Positioning

The system is designed as an academic, replicable research environment and a challenger model approach that is solely targeted at model validation. While it does take into account certain limitations of real-world applications, it is not a practical, working loan application system and it is not designed to replace any regulated institution's credit system.

Target users of the framework include credit-risk analysts creating challenger models, model-validation teams evaluating assumptions and data, and portfolio decision-makers distinguishing performance from reliability of predicted probability. If a bank is considering switching from a transparent scorecard to a more versatile ensemble model, decision-making cannot stop at whether the new model performs better but needs to account for cost-effective validation and governance as well. Alonso and Carbó (2020) describe this as the trade-off between prediction and supervisory cost.

The practical utility of the proposed framework consists in providing an audit trail from a documentable historic cohort to the out-of-time model comparison. An analyst may find out which loans qualify, what variables have been removed from the PD prediction, what transformations have been learned from training data, and performance dynamics in a following issue-year cohort. The scope is smaller than the commercial lending platform as no live application flow has been added, no outcomes for the rejected applicants provided, and no institutional policy enforcement or protected group monitoring has been included.

An understandable rationale exists for undertaking documented challenger analysis of this kind; however, there is no proof of market viability through customer interviews, cost analysis, and pilot program deployment. Hence, the competitive argument ought to be made on the basis of methodology: a transparent process that enables the testing of assumptions about validity is preferable to a model comparison which cannot be explained. It must not be made as an alternative to a proper lending system.



CHAPTER 2: THEORETICAL FRAMEWORK

The chapter introduces the theoretical and literature-based framework underlying the research. It discusses traditional credit scoring models, machine learning approaches, temporal validation, probability calibration, Expected Loss, explainability, fairness, and model governance. Finally, the chapter outlines the gap in methodology which this project aims to fill.

2.1. Traditional Models vs. Machine Learning in Credit Risk

The history of credit risk modeling is characterized by the constant search for improvement in terms of discrimination capabilities and operational stability. Earlier models like the Altman Z-score (Altman, 1968) relied on discriminant analysis to forecast bankruptcy of enterprises with the help of static financial ratios. Though the use of linear and conventional scorecards provided transparent and easily auditable criteria, their predictive capability decreased in application to high dimensional, non-linear datasets of consumer credit.

The transition from earlier methods to machine learning (ML) has been the breakthrough in the financial sector's capability to analyze large and heterogeneous data sets. Studies conducted by Khandani, Kim, and Lo (2010) and Lessmann et al. (2015) prove that ML algorithms perform better than traditional statistical approaches as they capture complex interaction between features of borrowers. Systematic literature reviews confirm the superiority of machine learning models, especially deep learning architectures, in terms of accuracy and predictive stability in credit scoring (Aboulmaouda et al., 2026; Chen, 2024; Dastile et al., 2020; Montevechi et al., 2024). Still, as Alonso and Carbó (2020) point out, organizations face a trade-off between the increase in predictive accuracy of complex algorithms and supervision cost increase. associated with "black-box" models.

Institutional scorecards and commercial enterprise credit engines, such as FICO, VantageScore, Zest AI, and Moody's Analytics RiskCalc, come with proven institutional deployment practices; however, they often require costly licenses and lack transparency (Alonso & Carbó, 2020). Conversely, the peer-reviewed academic literature pertaining to P2P lending has increasingly relied on tree-based ensemble methods; for example, Hadji Misheva et al. (2021) have used Lending Club data to evaluate the use of SHAP and LIME interpretability frameworks. The current research builds upon this prior literature and proposes an entirely reproducible and open-source workflow that stresses chronological out-of-time validation and leakage management to circumvent vendor lock-in.

2.2. The Rise of LightGBM in Tabular Financial Data

Tree-based ensemble methods have become the most popular among all traditional linear models and neural networks for solving machine learning problems with tabular credit data. The gradient boosting algorithms have surpassed all the alternatives, with the Gradient Boosting Decision Trees (GBDT) being the predominant solution. LightGBM (Ke et al., 2017) uses two novel techniques, namely, Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB); these techniques significantly speed up the computations and handle sparsity of financial data well.

Boosting is solved by iteratively minimizing some loss function. For binary classification tasks of default vs full repayment, the loss function is the log-loss function:

$$\text{Log Loss} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \quad (2)$$

where y_i is the class label and p_i is the predicted probability. Like other machine learning solutions for credit (Bazarbash, 2019; Bhuiyan & Sweet, 2024; Noriega et al., 2023), LightGBM's architecture demonstrates outstanding discrimination and is post-hoc interpretable.

2.3. Temporal Validation and Data Leakage

There are inherent dynamics in the credit population and outcome since the lending process involves changing criteria for approval, changes in the population of borrowers, and changes in the economic environment. In the assessment of models through random splitting of the data into training and test sets or through k-fold cross-validation, look-ahead bias tends to result in inflated estimates of the future performance of the model. For a robust credit modeling process, it is necessary to use out-of-time (OOT) validation that tests for longitudinal stability. A distinction needs to be made between pre-application features, which will include factors used in the application for approval, and post-application features such as recoveries and total amounts received. This is necessary to ensure that there is no data leakage as part of the model training.

2.4. Beyond Binary Classification: Class Imbalance, Calibration, & Expected Loss

One of the shortcomings in academic machine learning credit research pertains to the use of binary classifiers based on decision thresholds that are static. While metrics such as ROC-AUC can effectively evaluate whether a model has the capacity to order risks, they are inherently incapable of accounting for absolute probability accuracy. Thus, a model may correctly order the risks yet fail to estimate the default

rate adequately. Models used in the context of banking regulation according to the requirements of Basel III and Model Risk Management must provide sufficiently well-calibrated estimates of credit-event probabilities for applications that are dependent on probability levels rather than ranking alone.

It can be stated that the above-mentioned requirement directly fills one of the gaps discovered in the previous iterations of this research. In *Part I* (Arain, 2026), the issue of extreme class imbalance was solved with the help of synthetic minority over-sampling technique, or SMOTE (Chawla et al., 2002). While this method is beneficial when dealing with discrimination of minority classes, it makes the raw probability outputs unreliable for financial forecasting. For the transition from mathematical prediction to financial application, the models must be calibrated with methods such as Isotonic Regression. Such process is measured with Brier score that evaluates the mean squared difference between predicted probabilities (p_i) and actual outcomes (o_i):

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}(p_i - o_i)^2 \tag{3}$$

With the PD calibrated properly, PD acts as an important factor in calculating the expected loss, which is defined in Equation (1). In Equation (1), LGD stands for loss given default, and EAD stands for exposure at default. Using these factors, lenders are able to set the ideal decision-making thresholds with the goal of increasing the origination level without compromising their high-risk tolerance standards (Shi et al., 2022; Suhadolnik et al., 2023).

2.5. Explainable AI (XAI) in Finance

The adoption of non-linear models like LightGBM will need to satisfy the transparency requirement of the ECOA. Explainable AI (XAI) can help achieve a balance between improved predictive power and regulatory requirements. The SHAP framework (Lundberg & Lee, 2017), which uses game theory, can help explain the impact of each feature on the predicted results both globally and locally.

For any specific feature i, its Shapley Additive Explanations valueϕ_i is defined as follows:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!}[f(S \cup \{i\}) - f(S)] \tag{4}$$

In credit scoring, SHAP can allow for the identification of the most significant factors in model predictions and can contribute to testing the plausibility and appropriateness of these relationships. However, feature attributions do not provide evidence of economic causation or preclude the detection of

spurious relationships. SHAP explanations can facilitate the process of model governance and development of reason codes but cannot independently serve as the basis of adverse-action notices (Hadji Misheva et al., 2021; Pathi & Pothineni, 2025; SCALE D2C, 2026).

2.6. Algorithmic Fairness and Regulatory Evolution

As credit models become extremely optimized to reduce financial losses, there is a risk of embedding system-level biases via proxies. The concept of algorithmic fairness in machine learning often revolves around Equality of Opportunity (Hardt, Price, & Srebro, 2016), implying that all qualified applicants from various demographic or socio-economic segments would have an equal opportunity for being approved.

Within the realm of credit risk, this goal is measured through the metric of False Positive Rate (FPR), indicating the frequency of rejection of the eligible applicants. The definition of FPR is as follows:

$$FPR = \frac{FP}{FP + TN} \tag{5}$$

Error rate group-level differences, such as the False Positive Rate difference or ratio, are diagnostic measures of allocation difference in protected groups or their proxies. Such measures do not in themselves prove discriminatory or disparate treatment or non-compliance with anti-discrimination regulations. In the current study, renter/non-renter differences will therefore be considered proxy group diagnostics rather than as a formal protected-class fairness assessment.

2.7. Cross-Study Synthesis and Methodological Limitations

An analysis of literature shows a consistent trade-off between prediction optimization and operational feasibility. Despite the multiple comparative studies, which proved the superiority of non-linear ensembles (LightGBM, XGBoost, etc.) over traditional Logistic Regression on discrimination measures, such studies are often based on static datasets, random data splits, and metrics of accuracy instead of calibrated probabilities. Furthermore, explanations of the models are provided without creating any connections to regulatory governance or adverse action requirements. It is evident that the mere predictive performance cannot be the basis for deploying an algorithm; its performance on calibration, leakage prevention, and stability needs to be assessed as well.

2.8. Identified Research Gap

A survey of the literature shows a significant disparity between credit- modeling in machine learning and the demands of practical banking application. Many comparative analyses highlight the

importance of optimizing algorithms in terms of static and balanced datasets and evaluating their performance based on discrimination measures (Arain, 2026; OpenIntro, n.d.).

Modern finance literature implies that the proper credit model should cover multiple dimensions at once. The present thesis specifically aims to fill that multidimensional gap by converting the fundamental binary classification model of Arain (2026) into a cost-sensitive, drift-aware financial engine. Namely, it constructs a replicable and leakage-free framework that includes the following elements:

1. Clearly defined and resolved final states instead of fixed-horizon approximations.
2. Out-of-time (OOT) validation for assessing stability under temporal distribution shift across loan vintages.
3. Probability calibration tied to Expected Loss decision simulation.
4. Comparative analysis separating borrower/application properties from lender-generated pricing.
5. Explainability and fairness sensitivity analysis using socioeconomic proxies.

literature synthesis matrix

Approach	Discrimination	Probability calibration	Temporal validation	Explainability	Financial decision layer	Fairness/governance
Traditional scorecard / LR	✓	Possible	Often limited	High	Limited	High auditability
Tree-based ML	Strong	Requires assessment	Required	Post-hoc	Limited by itself	Requires governance
Typical academic credit classification	Primary focus	Variable	Often random split	Increasing	Often absent	Variable
Present framework	✓	✓	✓	✓	✓	✓

Comparative Analysis

Modelling approaches and their relevance to this project

Approach	Strengths	Weaknesses	Relevance to this project
Logistic Regression / scorecard	Transparent coefficients; easy benchmarking; familiar governance; relatively simple calibration	May miss nonlinear interactions; requires careful transformations and feature design	Primary interpretable benchmark
Random Forest / gradient boosting	Captures nonlinear effects and interactions; often strong tabular performance	More difficult to explain directly; can require calibration and tuning; may be unstable under drift	Candidate nonlinear comparison
Naive academic ML benchmark	Easy to implement; low software cost; useful for teaching	Often uses random splits, leakage-prone preprocessing, immature labels, and accuracy-focused metrics	The revised workflow is designed to correct these weaknesses
Commercial AI platforms	Integrated deployment, monitoring, governance, and experimentation tools	Licensing cost, vendor dependence, restricted transparency, and implementation complexity	Industry comparator, not a direct numerical benchmark
Internal institutional scorecard	Access to proprietary data, policies, protected-group monitoring, and operational outcomes	High development, governance, and data-integration requirements	More realistic production reference
Proposed framework	Reproducible open-source workflow; historical cohort selection; chronological validation; separate outcome table; explainability-ready architecture	Not production validated; no live data; no verified protected-group data; LGD/EAD not yet empirically modelled	Academic proof of concept and methodological framework

Table 2.1: Comparative Analysis of Credit Risk Frameworks and Commercial Platforms

SWOT Analysis

Credit-risk modelling frameworks

Framework	SWOT Category	Details
Traditional scorecard	**Strengths**	• High transparency. • Easy to document and challenge. • Familiar to credit-risk practitioners. • Suitable as a benchmark.
	Weaknesses	• May underrepresent nonlinear interactions. • Requires careful binning, transformations, and stability monitoring. • Can be less flexible for complex feature relationships.
	Opportunities	• Combine with monotonic constraints, scorecard development, and calibration. • Use as a challenger model against boosting.
	Threats	• May be outperformed on discrimination by well-governed ensemble models. • Can become outdated when borrower populations or economic conditions change.
Academic machine-learning benchmark	**Strengths**	• Rapid experimentation. • Broad algorithm comparison. • Low licensing cost. • Useful for methodological teaching.
	Weaknesses	• Random splits can overstate future performance. • Preprocessing can leak information. • Accuracy or F1 may be prioritised over PD calibration. • Static thresholds may ignore financial costs. • Explanations may be added after modelling without connecting them to governance.
	Opportunities	• Improve using out-of-time validation, calibration, leakage controls, and decision-cost analysis.
	Threats	• Misleading reported performance. • Poor reproducibility. • Inappropriate deployment or regulatory interpretation.
Commercial AI platform	**Strengths**	• Integrated experiment tracking, deployment, monitoring, and governance. • Potentially faster operationalisation. • Support for model comparison and workflow automation.
	Weaknesses	• Cost and vendor lock-in. • Proprietary components may limit independent auditability. • Outputs still depend on the institution's data, target, and validation design.
	Opportunities	• Used as an implementation environment after a sound research methodology has been established.
	Threats	• Automation can create false confidence. • Governance weaknesses may be hidden behind a polished interface.
Proposed framework	**Strengths**	• Uses 603,589 resolved 36-month loans rather than the earlier small snapshot. • Explicitly defines the target. • Uses chronological train, validation, and test periods. • Separates PD inputs from repayment outcomes. • Stores raw, processed, feature-rule, and preprocessing artifacts. • Compares a borrower/application baseline with a lender-pricing feature set. • Supports calibration, SHAP, and later Expected Loss analysis.
	Weaknesses	• Historical Lending Club data do not represent all lenders or all applicants. • The cohort contains originated loans only. • The target is final observed status, not a fixed-horizon PD. • No full protected-group fairness audit is currently possible. • LGD and EAD are not yet empirically validated. • The current system is a reproducible research pipeline, not a production lending platform.
	Opportunities	• Add calibrated PD models. • Compare Logistic Regression with LightGBM or another ensemble. • Add SHAP explanations. • Estimate or scenario-test LGD and EAD. • Add population-stability and performance-drift monitoring. • Add fairness analysis when appropriate governed data are available.
	Threats	• Label and cohort selection bias. • Vintage and economic drift. • Data-source changes. • Misinterpretation of illustrative fairness or Expected Loss assumptions. • Overclaiming regulatory compliance from explainability alone.

Table 2.2: SWOT Analysis of Industry Credit Models versus the Proposed Framework

CHAPTER 3: METHODOLOGY

3.1. Dataset Description and Cohort Selection

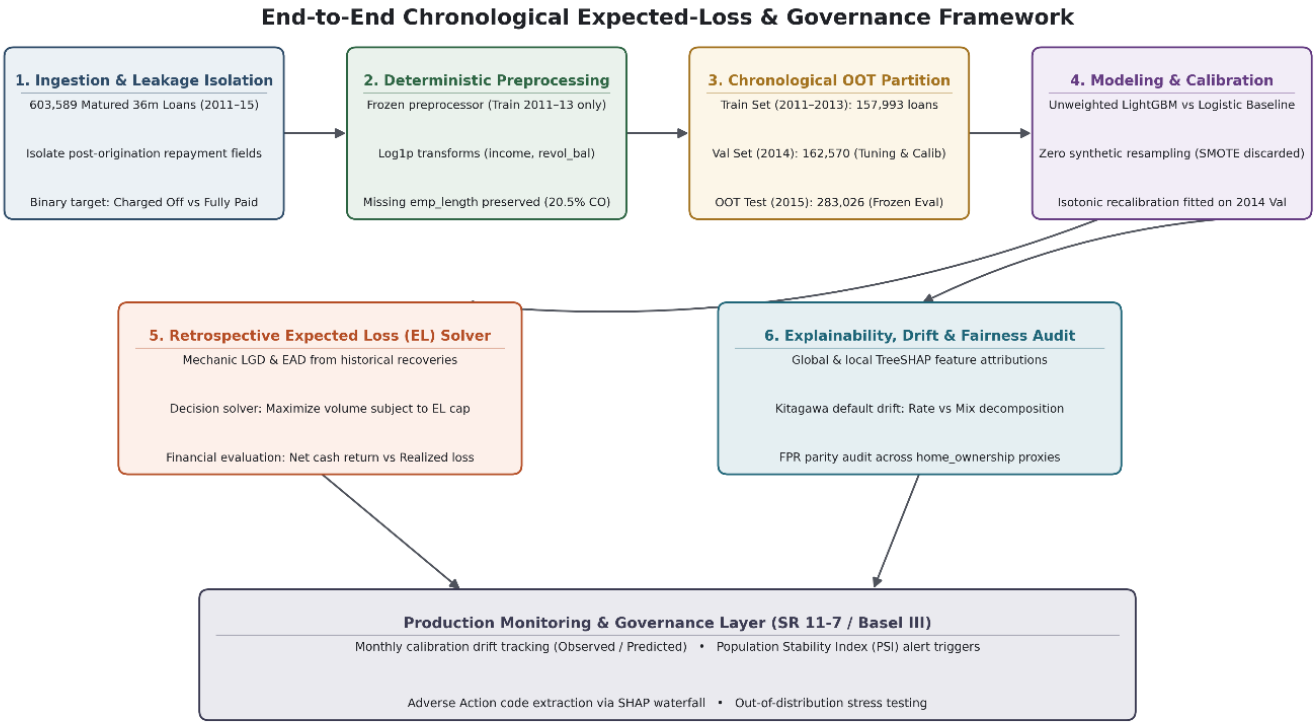


Figure 3.1: End-to-End Chronological Expected-Loss & Governance Architecture.

Figure 3.1 illustrates the macro-architecture of the Version 2.0 credit risk model framework, detailing the formal move from the exploratory, static classification pipeline of Part I (Arain, 2026) to a temporally validated, research-oriented credit-risk and financial decision framework. While standard machine learning performance benchmarking methods rely on synthetic resampling schemes (e.g., SMOTE), as well as random data split—methods that artificially shift the empirical priors, introducing a look-ahead bias (Chawla et al., 2002; Khandani et al., 2010), the pipeline is strict in imposing temporal separation between 603,589 matured 36-month loans[1]. Repayment artifacts post-origination are segregated upon ingestion to avoid target leakage. The feature engineering and transformation steps are estimated only

[1] Restricting to only originated loans with completely resolved 36-month outcomes results in a 'delayed label' setting. On a live production pipeline as of January 1, 2015, loans originated in late 2014 would still lack final repayment labels. A live implementation would necessitate using survival analysis or censoring techniques due to such immature labels.

on the 2011-2013 training vintage. The probabilistic predictions of the unweighted LightGBM model are then corrected via the Isotonic Regression mapping fitted on the 2014 validation cohort, before feeding into the Expected Loss Equation (1) constrained portfolio optimization step (Basel Committee on Banking Supervision, 2010; Ke et al., 2017). Lastly, decision artifacts on the out-of-time test cohort in 2015 are subjected to post-hoc SHAP explainability, Kitagawa drift decomposition, and ECOA socioeconomic proxy audit checks for fairness (Hardt et al., 2016; Lundberg & Lee, 2017).

For the empirical study, the data from the Lending Club accepted-loans dataset from 2007 until 2018Q4 has been utilized. In order to ensure that there is an actual mature credit portfolio, the sample of observations has been carefully filtered to include only 36-month loans originated between January 2011 and December 2015 with a status "Fully Paid" or "Charged Off". This resulted in a solid database of 603,589 resolved loans (519,362 Fully Paid and 84,227 Charged Off). Target variable `charged_off` was considered to be a binary variable, where 1 denotes charged off, and 0 – fully paid status. It was important that all post-origination data (e.g., total principal received, recoveries) was separated in the outcomes table to avoid data leakage.

Cohort-resolution audit. The raw 2018Q4 snapshot included an additional 147 36-month loans originated in 2015 that still lacked a recorded final outcome label: 72 were labeled as Current, 9 as In Grace Period, 5 were Late (16-30 days), and 61 were Late (31-120 days). Such loans were filtered out rather than labeled as non-charge-off. They comprise about 0.052% of 283,173 eligible loans originated in 2015 that are still subject to final-status filtering. The observed charge-off rate in the 283,026 fully resolved loans is 14.886%. If considering the 147 unresolved loans as future non-charge-offs as a conservative sensitivity bound, then the resulting 2015 charge-off rate would only be reduced to 14.879%, which constitutes only approximately 0.008 percentage point reduction. In contrast, considering the 147 loans as future charge-offs would result in an increase in the rate to approximately 14.930%. Hence, the filtering for resolved final status has no noticeable impact on the overall 2015 charge-off rate despite unknown future outcomes of these observations.

The need for a dynamic credit framework is demonstrated by analyzing the longitudinal changes in the Lending Club sample. From 2011 to 2015, there was rapid exponential growth in the loan origination amount. At the same time, there was a substantial temporal increase in the observed charge-off rate, which grew from around 10% in early 2011 to the point where the value was higher than 15% in late 2015. Thus, both growth in loan origination volume and increasing default risk make the static modeling invalid and require Out-of-Time (OOT) validation to ensure future stability.



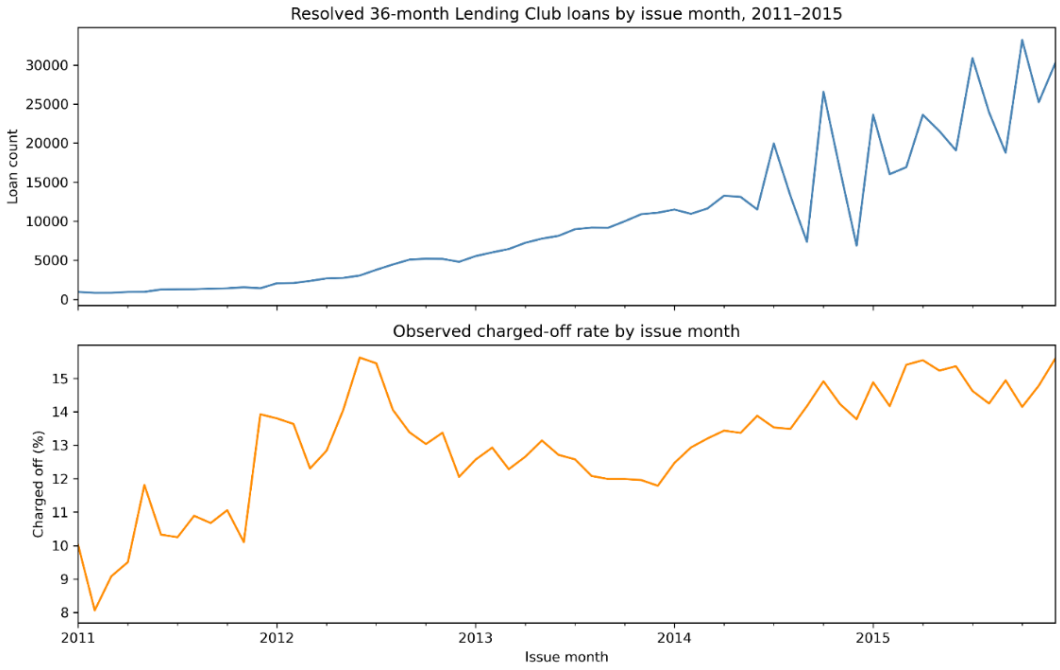


Figure 3.2: Longitudinal pattern of resolved 36-month loan volume and the corresponding observed charge-off rate throughout the 2011–2015 cohort, reflecting substantial target drift over the 60-month period.

3.1.1. Separation of Modelling Data and Payment Outcomes

To preserve the separation between information available for risk modelling and information observed only after loan origination, repayment outcomes were maintained in separate datasets. Notebook 01 created the primary PD cohort and a restricted loss-outcome table containing the fields required for principal-loss analysis. Notebook 01b subsequently extracted additional payment fields, including total payments, interest received, late fees and recovery-related amounts, for exactly the same 603,589 loans and in the same row order. These post-origination variables are used exclusively for retrospective financial evaluation and are never used as predictors or policy-decision inputs.

A reconciliation audit also established that `total_pymnt` already includes recorded recoveries for charged-off loans. Consequently, later net-cash calculations do not add recoveries a second time. This separation permits the later financial analysis to incorporate interest income while maintaining strict leakage control in the PD modelling pipeline.

3.2. Chronological Out-of-Time (OOT) Validation

In traditional credit risk research, the most popular approaches are random k-fold cross-validation and random train-test split. Such approaches can introduce look-ahead bias and may fail to capture temporal

distribution shift across loan vintages. To implement real-world restrictions and check the longitudinal performance, the OOT framework described below has been implemented:

- **Training Set (2011–2013):** 157,993 loans used for model fitting and creation of the preprocessing pipeline.
- **Validation Set (2014):** 162,570 loans used for out-of-sample model comparison, probability recalibration, and subsequent development-stage policy choices.
- **Held-Out Test Set (2015):** 283,026 loans strictly reserved for final, out-of-time performance and financial impact evaluation.

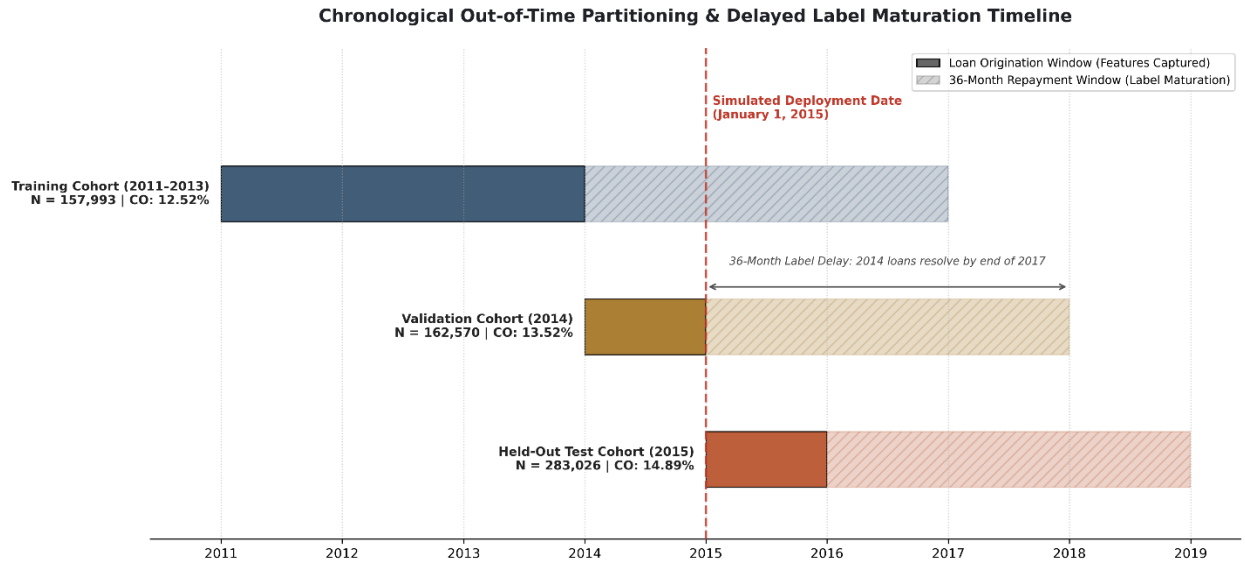


Figure 3.3: Chronological Vintage Maturation and Out-of-Time Validation Timeline.

Figure 3.3 illustrates the longitudinal structure of the empirical cohort alongside the 36-month label maturation schedule. In consumer credit underwriting, standard cross-validation is invalid because of longitudinal changes in broader economic conditions, portfolio seasoning, borrower composition, and underwriting practices (Federal Reserve, 2011; Lessmann et al., 2015). By breaking the 603,589 loan samples in chronological order, training models on the 2011–2013 vintages ($N = 157{,}993$), conducting validation and probability recalibration on 2014 ($N = 162{,}570$), and reserving 2015 for final OOT evaluation ($N = 283{,}026$), one avoids contamination with any forward-looking information. At the same time, the timeline shows a structural "delayed label" issue that was discussed in Section 1.6: due to the fixed-term structure of 36-month loans, three full years of time are needed for charge-offs and repayments to become resolved. This implies that the 2014 vintages were resolved by December 2017 only. Consequently, simulating a production launch in January 2015 based on fully resolved validation labels

produces retrospective benchmark instead of actual production feed, underscoring the importance of ongoing monitoring to tackle the problem of post-launch calibration drift.

In order to account for distributional drift between loan vintages, probability recalibration through Isotonic Regression was performed only on the 2014 validation sample. To characterize distributional shift in the loan population between loan vintages, the Population Stability Index (PSI) is used to quantify distributional changes in model-inputs, whereas Kitagawa decomposition decomposes observed changes in the charge-off rate between populations into portfolio-mix and within segment effects.

3.3. Data Preprocessing and Feature Engineering

A reproducible baseline feature policy was created based on the information available at origination from the borrower, application, and credit history datasets. While the initial dataset extraction had found 22 baseline columns, the finalized frozen modeling pipeline had used 21 engineered input features (15 numerical and 6 categorical)[2]. Five out of those 22 baseline columns (`earliest_cr_line`, `annual_inc`, `revol_bal`, `fico_range_low`, and `fico_range_high`) were dropped and replaced with 4 engineered versions of these columns (`credit_history_months`, `annual_inc_log`, `revol_bal_log`, and `fico_midpoint`).

Feature Name	Type	Transformation	Imputation	Baseline Included	Excluded in Sensitivity Variant(s)
`loan_amnt`	Numerical	Standard Scaled	Median	Yes	None
`annual_inc_log`	Numerical	log1p, Standard Scaled	Median	Yes	None
`dti`	Numerical	Standard Scaled	Median	Yes	None
`delinq_2yrs`	Numerical	Standard Scaled	Median	Yes	None

[2] Although the date of loan origination (`issue_month` or `issue_d`) is strictly excluded from the final set of input features to avoid temporal data leakage and prevent the algorithm from memorizing the information—effectively using the calendar time variable as a surrogate for cohort-specific risk level—it must be used during the preprocessing stage. Namely, it serves as an important chronologic reference to calculate the `credit_history_months` feature, which measures how much time passed since the borrower got his/her `earliest_cr_line`.

Feature Name	Type	Transformation	Imputation	Baseline Included	Excluded in Sensitivity Variant(s)
`inq_last_6mths`	Numerical	Standard Scaled	Median	Yes	None
`open_acc`	Numerical	Standard Scaled	Median	Yes	None
`pub_rec`	Numerical	Standard Scaled	Median	Yes	None
`revol_bal_log`	Numerical	log1p, Standard Scaled	Median	Yes	None
`revol_util`	Numerical	Standard Scaled	Median	Yes	None
`total_acc`	Numerical	Standard Scaled	Median	Yes	None
`collections_12_mths_ex_med`	Numerical	Standard Scaled	Median	Yes	None
`acc_now_delinq`	Numerical	Standard Scaled	Median	Yes	None
`pub_rec_bankruptcies`	Numerical	Standard Scaled	Median	Yes	None
`credit_history_months`	Numerical	Derived (from date), Scaled	Median	Yes	None
`fico_midpoint`	Numerical	Derived (averaged), Scaled	Median	Yes	None
`emp_length`	Categorical	One-Hot Encoded	Explicit "Missing"	Yes	None



Feature Name	Type	Transformation	Imputation	Baseline Included	Excluded in Sensitivity Variant(s)
`home_ownership`	Categorical	One-Hot Encoded	None	Yes	`no_home_ownership`, `no_proxy_trio`
`verification_status`[3]	Categorical	One-Hot Encoded	None	Yes	`no_verification_status`, `no_proxy_trio`
`purpose`	Categorical	One-Hot Encoded	None	Yes	None
`application_type`	Categorical	One-Hot Encoded	None	Yes	None
`addr_state`	Categorical	One-Hot Encoded	None	Yes	`no_addr_state`, `no_proxy_trio`

Table 3.1: 15 numerical and 6 categorical features, summarizing the pipeline.

- **Deterministic Transformations:** The missing employment length variable (`emp_length`) was retained as a "Missing" value as opposed to being imputed because exploratory data analysis indicated high predictability of that missingness, as well as its association with 20.48% charge-offs. Highly skewed features, in particular annual income and revolving balance, were subject to a $\log(1 + x)$ transform.
- **Pipeline Engineering:** A preprocessor was trained solely on the 2011–2013 cohort. Numerical features underwent a median imputation and standard scaling, while categorical features were processed through a One-Hot Encoding scheme. The pipeline was configured to ignore new unknown categorical values in 2014 and 2015 datasets.

Exploratory Data Analysis of the numerical features in the 2011–2013 training cohort determined the need for certain deterministic transforms before applying any machine learning algorithm. Features such as `annual income`, in particular, showed heavy right-skew, which tends to negatively affect the performance of gradient-based algorithms. The transformation of income using a $\log(1 + x)$ successfully

[3] The operational presence of `verification_status` before the credit decision was made is unverified in this dataset. Moreover, `verification_status` is often triggered by the risk assessment performed by the lender, thus acting as a partial risk score in addition to being an independent applicant characteristic.

fixed the skew and made it easier to separate the fully paid from the charged-off loans. At the same time, features such as `DTI` and `Revolving Utilization` (capped at the 99th percentile for illustration purposes) have more gradual and overlapping distributions., underscoring the necessity of non-linear tree ensembles capable of capturing complex boundary interactions.

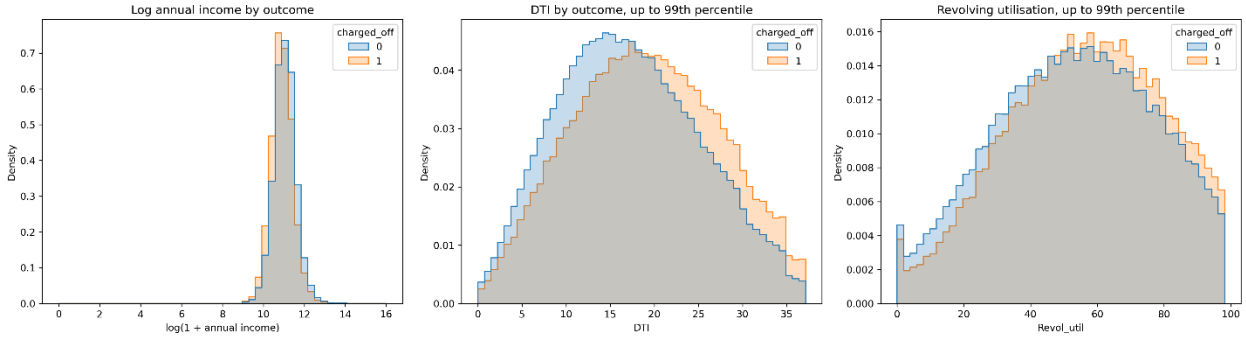


Figure 3.4: The density distributions of the Log Annual Income, Debt-to-Income Ratio (DTI), and Revolving Utilization according to the loan outcome classes are considered. The logarithm transformation corrects the skewness of income, while the DTI and Revolving Utilization provide some insights into the overlapping, non-linear risk boundaries among the classes.

3.4. Comparative Modeling Design

In order to compare the effectiveness of non-linear models with traditional banking methods, the framework uses a matrix of comparisons using chronological splits:

- **Algorithm Comparison:** The Logistic Regression model is used as a clear, understandable baseline, while the unweighted LightGBM is used as a challenger.
- **Probability Output:** Raw model probabilities are explicitly compared to isotonic calibrated probabilities.
- **Feature Policies:** The baseline borrower/application features are compared to multiple sensitivities (for example, the omission of socioeconomic/geographic proxies).
- **Lender-Pricing Reference:** An additional model that includes the Lending Club's proprietary grades, sub-grades, and interest rate features was built. However, it serves only as an upper bound of the comparison and not as a fair challenger since the price features are a function of the underwriting process of the lender after the application.

- **Financial Evaluation:** Finally, the frozen Expected Loss acceptance policy is compared to an "accept all" baseline decision using the sequential evaluation based on the realized Principal Loss and undiscounted net-cash return.

Unlike earlier iterations of the research (Arain, 2026), the synthetic resampling algorithms such as SMOTE were left out. While comparing the results of using the unweighted and the `class_weight='balanced'` parameters, the unweighted LightGBM model was chosen as the main challenger since the balanced weight had heavily skewed raw probability outputs increasing the Brier Score error by two times (0.2309 for the balanced variant and 0.1140 for the unweighted variant).

No hyperparameter search was performed for the LightGBM challenger. The reported specification uses 200 estimators and a learning rate of 0.05; therefore, the study evaluates a fixed challenger specification rather than claiming an optimized LightGBM model.

3.5. Probability Calibration

One of the goals of the modelling framework is not only the sorting of borrowers based on their relative risk but also the validation of the predicted probabilities in light of the actual charge-off rates. For the validation group of 2014, the raw unweighted LightGBM had a mean charge-off probability of 13.05%, in contrast to the actual charge-off rate of 13.73%. The corresponding mean prediction for Logistic Regression was 13.26%. Decile-based reliability plots showed that both models matched the actual risk fairly well, showing underprediction on the aggregate level.

Isotonic regression was used as a pre-specified recalibration step, but not necessarily as one that was supposed to be applied to improve future results. The LightGBM model was trained on the training cohort from 2011–2013, while only the mapping procedure was performed on the 2014 sample. The mapping itself was used without any modifications for the 2015 held-out sample. The obtained probabilities can thus be seen as a calibrated estimate of the actual charge-off of the loan based on the final status of the loan.

The major caveat here is the fact that complete final statuses of loans originated in 2014 would not have been available in a live environment at the beginning of 2015. Thus, the recalibration exercise for 2014 can be treated as a OOT benchmark, but not as an accurate reflection of January 2015 production process.



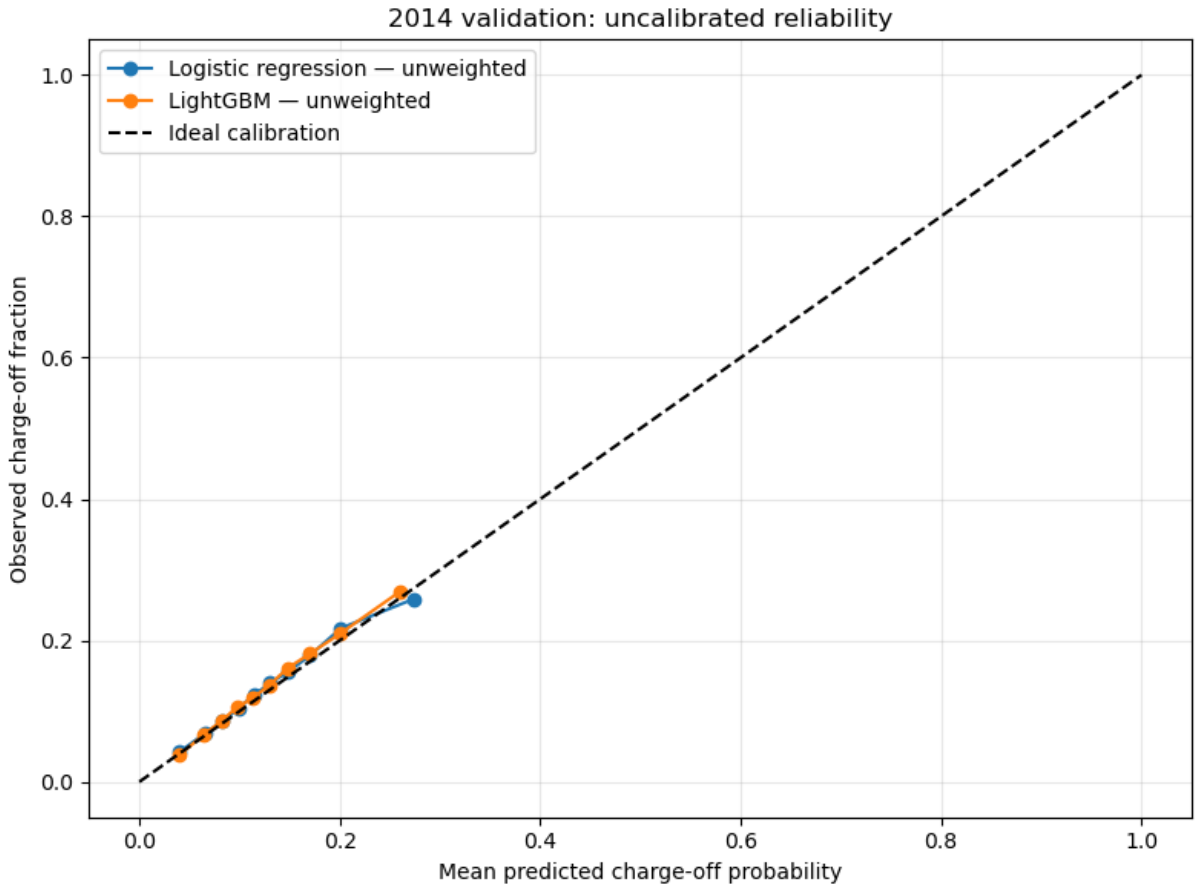


Figure 3.5: Un-calibrated reliability plot for the 2014 Validation dataset. Both the Logistic Regression and LightGBM models are below the ideal calibration line, which indicates systematic under-prediction of true charge-offs before isotonic mapping.

CHAPTER 4: RESULTS

4.1. Out-of-Time Model Discrimination

On the 2015 OOT cohort, the raw LightGBM challenger achieved higher discrimination than Logistic Regression across ROC-AUC, Gini and KS. The difference in ROC-AUC was +0.0057, with a 95% bootstrap confidence interval of [+0.0045, +0.0072]. Because this interval excludes zero, the difference is statistically distinguishable within the test cohort, although its absolute magnitude remains modest.

Model	ROC-AUC	Gini Coefficient	KS Statistic	95% AUC Confidence Interval
Logistic Regression (Baseline)	0.6569	0.3139	0.2253	[0.6545, 0.6596]
LightGBM (Raw Challenger)	0.6625	0.3251	0.2362	[0.6598, 0.6655]
LightGBM (Isotonic Calibration)	0.6624	0.3248	0.2360	[0.6597, 0.6654]

Table 4.1: Discrimination Metrics on 2015 OOT Test Set

Note: Isotonic Isotonic regression applies a monotonic non-decreasing transformation. It generally preserves score ordering but may introduce ties, which explains the very small change in ROC-AUC after calibration..

4.2. Probability Calibration Diagnostics

While being better in discrimination, the raw LightGBM model tended to under-predict the absolute level of risk in the 2015 cohort. The average prediction made by the raw LightGBM model resulted in a probability of 12.87% vs. 14.89% (observed), yielding an observed over predicted ratio of 1.1570.

We applied isotonic regression to calibrate the score of the model to match the empirical probability of defaulting observed in the 2014 validation cohort. As the 2015 cohort has experienced an even further increase in charge-off rate, this calibration has produced slightly underestimated probability of default in 2015 out-of-time cohort. This calibration has only marginally decreased the Brier Score error (from 0.1219

for raw to 0.1216 for isotonic). Despite this mathematical reduction in error, the 2014-fitted mapping could not fully correct the severe upward level-shift in actual defaults that continued through 2015.

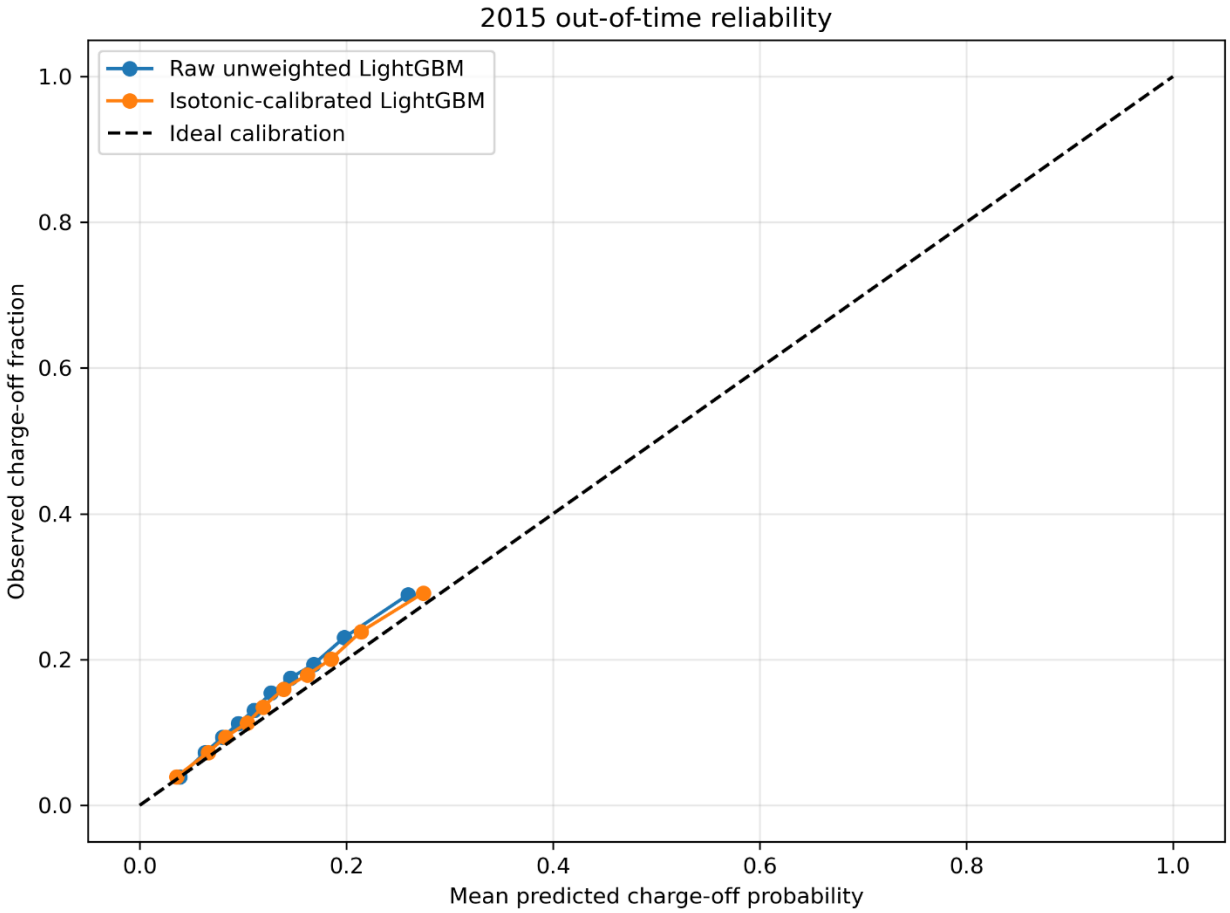


Figure 4.1: Reliability diagram comparing raw uncalibrated and isotonic-recalibrated LightGBM probabilities on the 2015 out-of-time test set.

However, when tested against the 2015 Out-of-Time test dataset, the calibration limits that are associated with historical data were evident. As the observed charge-off rate/risk level continued to increase in the 2015 cohort, the calibration model trained using 2014 data could not sufficiently account for the level shift. The comparison of raw LightGBM probabilities with Isotonic and Platt (H2) recalibration clearly indicates that, despite some progress in terms of improvement in mathematics, all models continued under-predicting the actual 2015 risk at higher probability deciles.

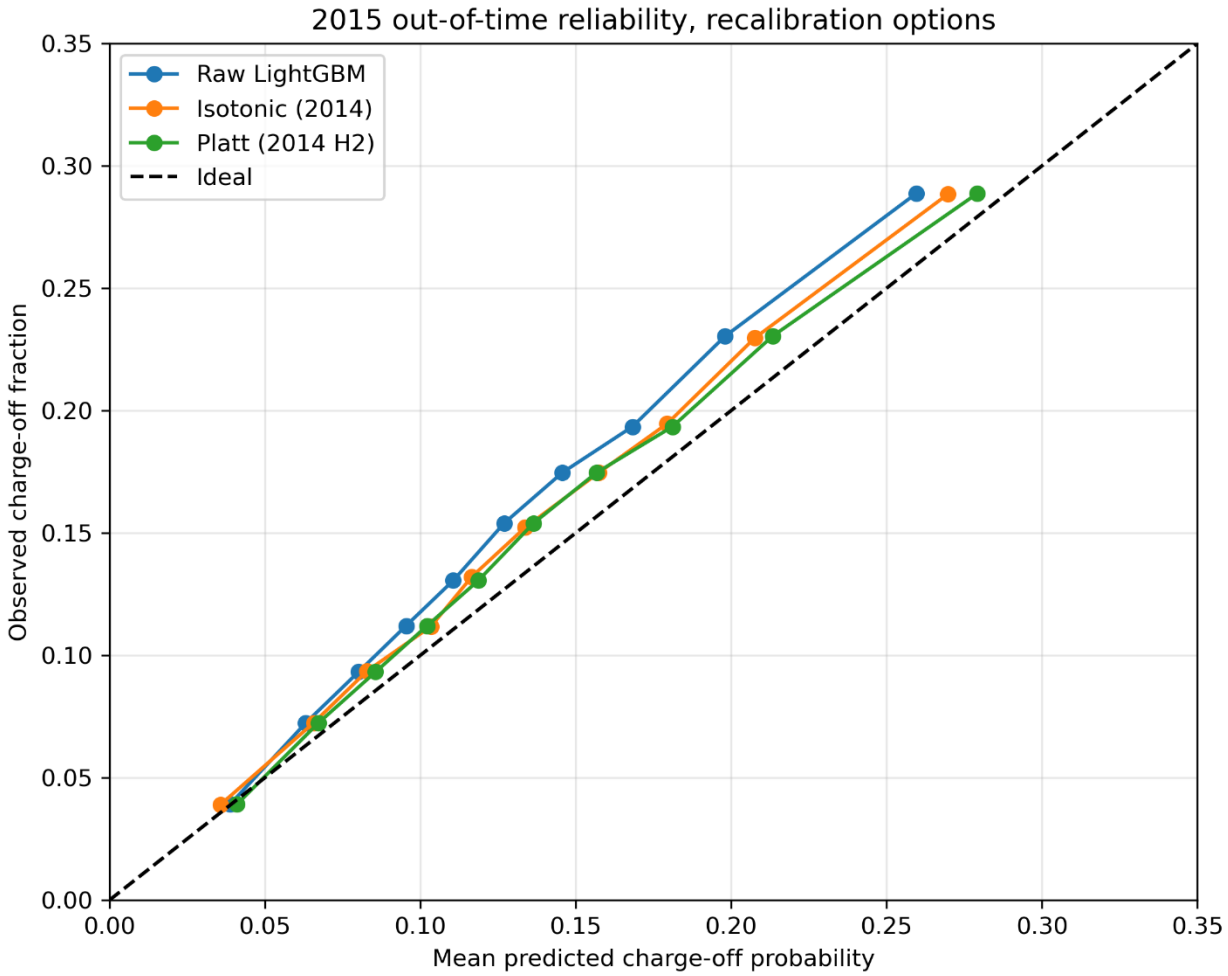


Figure 4.2: 2015 Out-of-Time reliability diagram: Raw LightGBM predictions vs. isotonic and Platt recalibration. Even with the advancements achieved mathematically, all three recalibrations trained on 2014 data under-predict the charge-off rate/risk level in 2015 cohort.

4.2.1. Formal Calibration Diagnostics and Alternative Recalibration

Moreover, the reliability diagrams have been enhanced with some formal diagnostics for calibration. For the 2015 OOT sample, the raw LightGBM had calibration slope equal to 1.0160 and calibration-in-the-large 0.1762. After using the isotonic mapping trained on 2014 data, the slope became equal to 0.9912 and the calibration-in-the-large was reduced to 0.1167. As both slopes are close to unity and the latter is still positive, the main problem with OOT calibration concerned the overall risk level change rather than the significant difference in dispersion of predicted risks.

Isotonic regression decreased the 2015 Brier score from 0.1219 to 0.1216 and the observed-to-predicted ratio from 1.1570 to 1.1008. However, isotonic regression failed to demonstrate any additional

improvement as compared to simpler recalibration techniques estimated on the 2014 data[4]. Both intercept shift and Platt recalibration provided the Brier score of approximately 0.1216 and log loss of 0.4001. The lowest log loss 0.3998 was obtained by Platt recalibration based on the second half of 2014 data. Hence, recalibration improved probability level alignment. However, the specific contribution of isotonic regression was not revealed.

4.3. Population Stability and Drift Decomposition

4.3.1. Population Stability of Model Inputs and Scores

In order to measure the changes in distribution that deteriorated the model performance, the Population Stability Index (PSI) was calculated for each feature in the dataset. PSI measures the degree of distribution change across both continuous and categorical features as follows:

$$\text{PSI} = \sum (\text{Actual \%} - \text{Expected \%}) \times \ln\left(\frac{\text{Actual \%}}{\text{Expected \%}}\right) \tag{6}$$

The Population Stability Index (PSI) has been used as a descriptive metric of distributional changes and not as a statistical test or cause-and-effect diagnostic tool. Most of the inputs for the baseline model had PSI lower than the conventional threshold of 0.10 when comparing the 2011–2013 training sample with 2015 test sample. The only exception was `verification_status` with PSI = 0.1805, indicating medium distributional shift. DTI variable was relatively stable with PSI = 0.0638. Distribution of the raw LightGBM score has also been found to be quite stable between 2014 validation period and 2015 test period with PSI = 0.0013.

In terms of comparative contextual analysis, the interest-rate variable in Lending Club obtained a population stability index (PSI) value of 0.2945 upon comparison between the 2011-2013 training cohort and the 2015 out-of-sample (OOT) cohort. It is important to note, however, that the `int_rate` variable is not a baseline input, but an output of lender pricing. Its high PSI therefore indicates substantial change in the observed pricing distribution rather than direct instability in the baseline model's input population.

The distributional stability of all 21 inputs of the baseline model between the 2011-2013 training cohort and the 2015 OOT test cohort is visualized in Figure 4.3 below. Grade and interest rate variables provided by the lenders are shown separately because they are contextual variables and not inputs to the fixed baseline model.

[4] Isotonic Regression performs the strictly monotonic, non-decreasing mathematical transformation. Thus, by definition, it fully preserves the original rank-order of probabilities returned by the classifier. As a result, its only effect is marginal change in ROC-AUC.

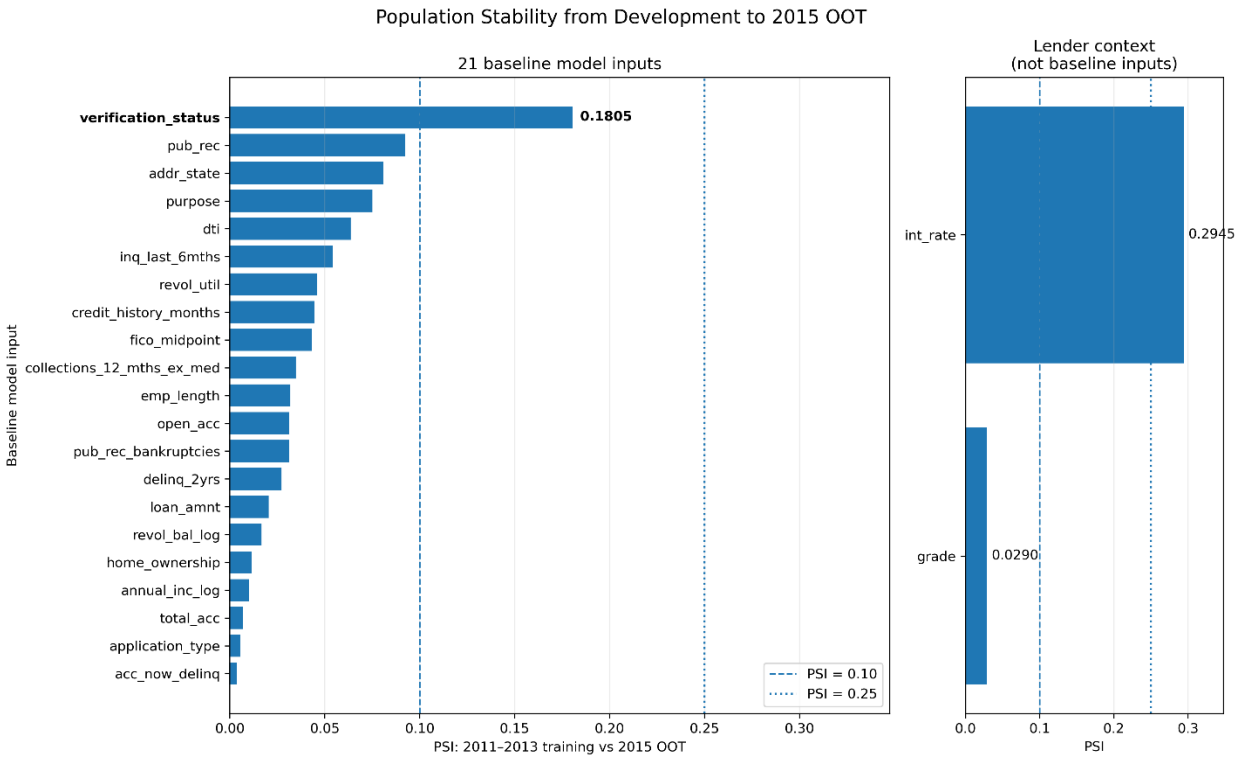


Figure 4.3 Population Stability Index (PSI) of the 21 baseline model inputs compared between the 2011-2013 training cohort and the 2015 OOT test cohort. Lender-provided grade and interest rate are separated as contextual variables since they are not inputs to the baseline model. The reference lines of PSI values 0.10 and 0.25 are marked as dotted and dashed, correspondingly.

From the figure above, it becomes clear that the stability of model inputs is different from stability in the general underwriting environment of the lender. The only input in the baseline model that exceeds PSI = 0.10 is the `verification_status` feature. Other model inputs have relatively low distributional changes, while the contextual interest rate variable is above 0.25. Given that the interest rate is outside the baseline feature policy, its change cannot be interpreted as the instability of the model input population. The stability of the predicted scores combined with the observed degradation of the charge-off rates leads to the further decomposition of the total effect.

4.3.2. Drift Decomposition

In order to find out the cause behind this increase in the default rate, the Kitagawa decomposition method was applied to split the drift into the "Mix Effect" (shifting toward inherently riskier borrowers) and a "Rate Effect" (identical borrowers defaulting more frequently):

$$\Delta = \sum (w_{new} - w_{base}) \cdot r_{base} + \sum w_{new} \cdot (r_{new} - r_{base}) \quad (7)$$

Accordingly, the decomposition shows that the increase in the observed charge-off rates was largely due to the deterioration in segments rather than the transition towards riskier segments. With 2014 LightGBM score deciles, the 1.160 percentage-point increase between 2014 and 2015 can be decomposed into a -0.203 pp mix effect and a +1.363 pp rate effect. In other words, borrowers living in comparable model-score segments exhibited higher charge-off rates in 2015. It is consistent with the observed deterioration in probability calibration, though the decomposition method per se cannot explain the causality behind.

Segmentation	Period comparison	Total change	Min effect	Rate Effect
Lending Club Grade	2011–2013 → 2015	+2.367 pp	-0.040 pp	+2.407 pp
Lending Club Grade	2014 → 2015	+1.160 pp	-0.576 pp	+1.735 pp
Model Score Deciles	2014 → 2015	+1.160 pp	-0.203 pp	+1.363 pp

Table 4.2: Kitagawa Decomposition of Observed Charge-Off Rate Changes Across Alternative Segmentations and Comparison Periods.

Furthermore, the sensitivity of the results to unresolved-loans reported in Section 3.1 shows that the 2018Q4 cutoff of observation period cannot be a reason for 1.160 percentage point difference in aggregate rates between 2014 and 2015: even with the conservative assumption that all 147 unresolved cases in 2015 will be non-charge-offs, the rate of 2015 is decreased by only 0.008 percentage points. Since the outcome of those loans is still not known, they are not considered as resolved cases within the Kitagawa decomposition.

Population Stability Index (PSI) diagnostics for inputs shows that core demographic attributes have been stable (for example, Debt-to-Income PSI is equal to 0.0638), and that lender pricing has changed (PSI is equal to 0.2945). Besides, the Kitagawa drift decomposition demonstrates that an increase in the default rate from 12.52% to 14.89% occurred because of internal deterioration (+2.407 percentage points) and not because of a poor shift in borrower portfolio mix.

4.3.3. Monthly Calibration and Discrimination Drift

The monthly analysis provides an extra insight that discrimination and calibration might change independently over time. The monthly ROC-AUC of the raw LightGBM was roughly steady across 2014–

2015, fluctuating around the mid-0.60s. At the same time, the ratio of the observed rate to the predicted one was increasing throughout 2015. In particular, the ratio of the raw LightGBM reached 1.2197 in December 2015. As for the 2014-fitted isotonic model, its ratio was 1.1615. Thus, the model continued ranking the loans with comparable efficiency while progressively underestimating their actual charge-off risks.

Note that the 2014 monthly values of isotonic model should be viewed with care, since isotonic mapping itself was built using all of 2014 validation outcomes. The 2015 monthly observations represent the true out-of-time (OOT) assessment of the frozen calibration mapping.

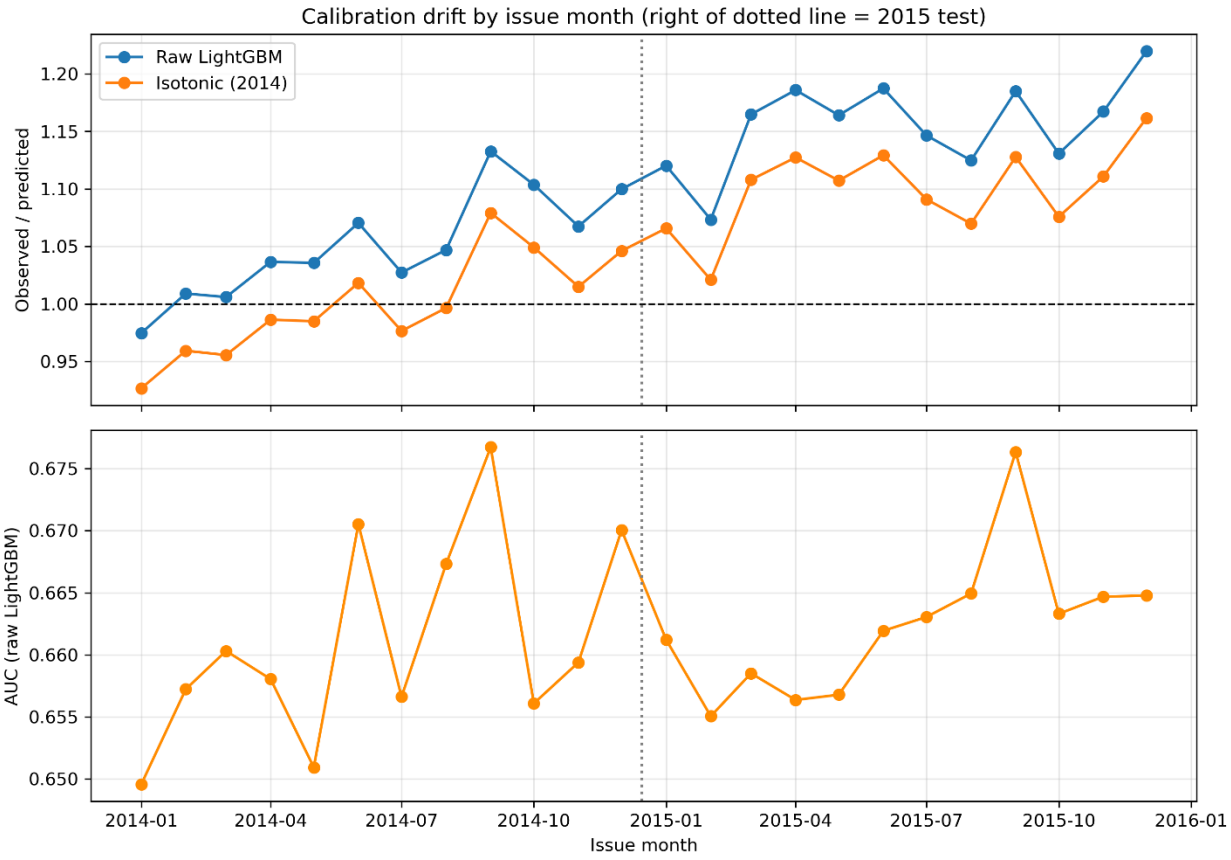


Figure 4.4: Monthly tracking of drift in calibration (upper graph) and stability of discrimination (lower graph) during the Out-of-Time period from 2014 to 2015. Though the value of the AUC remained broadly stable throughout the Out-of-Time period, the ratio of Observed/Predicted is gradually increasing.

4.4. Sensitivity Analysis and Fairness Implications

With motivations of fair lending and model governance concerns in mind, a feature-policy sensitivity analysis was done to assess the effect of excluding certain socioeconomic, geographical or operationally uncertain features on predictive performance as well as proxy group outcomes. Since the Lending Club dataset in question lacks verified protected class information, this sensitivity exercise can be

classified as a proxy-group sensitivity analysis and cannot be deemed as a disparate impact analysis. For each of the LightGBM feature-policy variations, the approval cut-off point was established based on 2014 scores such that approval rate in the development period would correspond to the baseline policy, which was then used unchanged in 2015. Matching the approval volumes in the development period would provide the feature policies under consideration with a more similar operational setting while not imposing identical realized 2015 loss rates and group outcomes.

4.4.1. Feature-Removal Sensitivity

Variant	2015 AUC	Principal-loss rate	Approval ratio: renter/non-renter	FPR ratio: renter/non-renter
Baseline	0.6625	5.27%	0.7521	1.8894
No verification status	0.6624	5.27%	0.7508	1.8931
No state	0.6617	5.24%	0.7505	1.8976
No home ownership	0.6593	5.32%	0.8327	1.4977
No proxy trio	0.6581	5.32%	0.8453	1.4497

Table 4.3: Feature Policy Sensitivity Outcomes (2015 OOT Test Set)

Excluding `verification_status` and `addr_state` had negligible effects on ranking performance and the renter/non-renter disparity measures. Exclusion of the `home_ownership` resulted in small decrease in 2015 AUC from 0.6625 to 0.6593 along with the FPR ratio drop from 1.8894 to 1.4977. At the same time, renter-to-non-renter approval ratio increased from 0.7521 to 0.8327. Simultaneous exclusion of all three proxy-sensitive features provided a slightly decreased AUC value of 0.6581, an FPR ratio of 1.4497 and an approval ratio of 0.8453. In summary, this demonstrates the sensitivity of group outcomes to the feature-policies but cannot be deemed as proof of causality, discrimination and illegality of any particular feature.

4.4.2. Structural and Reference Sensitivities

The two additional experiments help to highlight potential limitations beyond removal of direct features. In particular, the `delayed_labels_2011_2012` variant limits training to 2011-2012 to provide a rough sensitivity test of the 36-month outcome maturity delay. 2015 AUC goes down to 0.6459 from 0.6625, and the actual funded-principal loss rate goes up from 5.27% to 5.48%. The experiment shows that limitations of label availability could have a significant impact on the estimation of the model performance; however, it is just a rough approximation of a real point-in-time label availability design.

Second, the models with reference to lender data serve to show the usefulness of the underwriting output provided by Lending Club. The `pricing_reference` reaches the 2015 LightGBM AUC of 0.6831, while the `grade_only_reference` reaches 0.6784 compared with 0.6625 for the 21-feature application data baseline. The reference models are not considered as independent pre-approval challengers since the grade and pricing variables are generated in Lending Club's underwriting process. Their better performance indicates that there is predictive information in the lender underwriting outputs which was not captured in the selected application data baseline.

Results of the multidimensional sensitivity analysis are summarized in Figure 4.5, which shows 2015 out-of-time (OOT) discrimination versus the renter versus non-renter false positive-rate (FPR) ratio. This plot allows disentangling the prediction ranking performance from the proxy group disparity measure and helps to understand the trade-offs between the alternative feature policy choices. Principal loss realized results continue to be reported separately in Table 4.3.

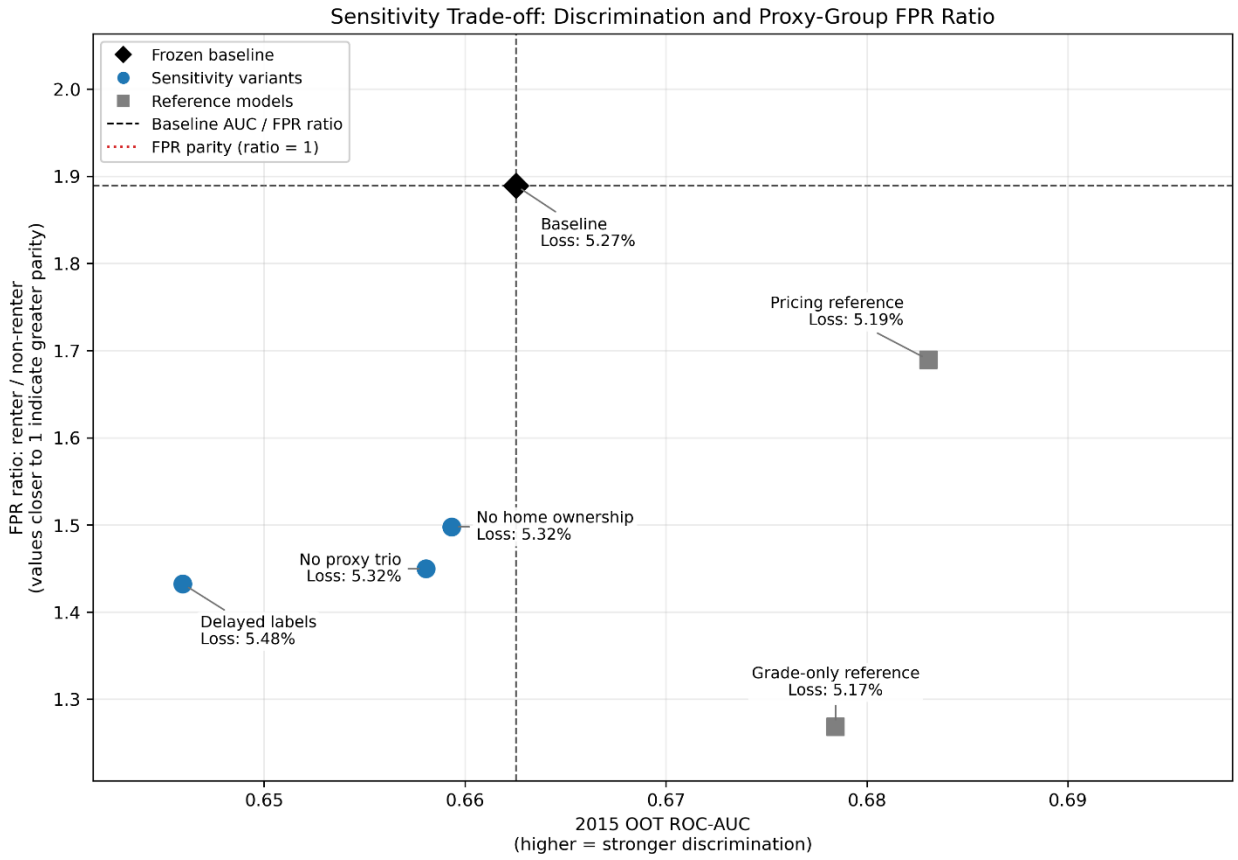


Figure 4.5: Trade-off between 2015 OOT discrimination and renter/non-renter FPR ratio for selected LightGBM sensitivity and reference variants. Dashed lines indicate the frozen baseline performance, while the FPR ratio of 1 implies proxy group parity. Realized funded-principal loss rates are included in the point annotations. Pricing and grade-only specifications are reference models and should not be interpreted as independent pre-approval alternatives.

The figure shows how the effects of policy feature changes are multifaceted. Dropping the feature `home_ownership` or all the three proxy sensitive features results in a large reduction in renter/non-renter FPR ratio towards unity while resulting in a small drop in ROC-AUC value. The delayed label method also results in a lower FPR ratio; yet, this is associated with a much greater drop in discrimination. In other words, the improvement of the group fairness measure in this case cannot be evaluated independently from weaker predictive discrimination achieved. Pricing and grade-only reference models yield greater discrimination; however, they rely on lender's underwriting data and thus answer a different modeling question than the application data one.

4.4.3. Interpretation Limits

The results for fairness sensitivity analysis should be treated with caution. Dropping a feature does not necessarily lead to eliminating the information associated with it since the correlated predictors income,

FICO, DTI, and loan purpose can preserve some of this information. Moreover, the renter/non-renter comparison was performed on the basis of socioeconomic proxy rather than a protected class. Therefore, the reported approval ratio and FPR ratio should be viewed as governance metrics and not a proof of fair lending compliance.

The feature-policy comparisons are performed at the same approval volume within the development period rather than at the same probability cutoff since the alternative models have different score scales. Moreover, the structural sensitivity methods use raw model scores without separate re-calibration of every model; therefore, their calibration should not be taken as directly comparative probabilities.

4.5. Retrospective Expected Loss and Portfolio Optimization

Whereas classic benchmarks of machine learning models focus on the accuracy of classification, the application of credit modeling implies dynamic decision-making based on real finances (Alonso & Carbó, 2020). In order to move from a probabilistic ranker model to a cost-sensitive underwriting process, the following Expected Loss (EL) framework was developed.

The framework uses the original funded amount (`funded_amnt`) as the proxy for Exposure at Default (EAD)[5]. The Loss Given Default (LGD) was calculated mechanically based on historical repayment results in 2011-2014 in order not to use data from 2015 test set as input. The principal loss fractions in charged off loans are equal to:

$$LGD_{raw} = 1 - \frac{\text{Total Principal Received} + \text{Recoveries}}{\text{Funded Amount}} \tag{8}$$

The values were strictly restricted to $[0,1]$, so that the exposure-weighted historical LGD assumption became equal to 50.0541%. An accounting anomaly was detected among historical charged off loans – there were 294 loans with negative raw principal-loss fractions, where the recorded principal received plus recoveries exceeded the original funded amount. Instead of taking them as negative credit losses, the principal-loss fractions were restricted to zero. As a result, the exposure-weighted LGD assumption becomes equal to 50.0541%; without including these observations, the exposure-weighted LGD assumption becomes equal to 50.4015%. The reason for such anomalies in historical data could not be found and can be considered as a limitation in measurements. With LGD and EAD defined, it becomes

[5] The original funded amount is used as a simplified ex-ante proxy for EAD. In the live production environment, EAD will require its own predictive sub-model for estimating the exact outstanding principal and drawn commitment at the time of default.

possible to calculate the Expected Loss for each loan using the isotonic-calibrated Probability of Default (PD):

$$EL_i = PD_i^{cal} \times \overline{LGD} \times EAD_i^{proxy} \tag{9}$$

$$R_{EL}(t) = \frac{\sum_{i:PD_i \le t} E\,L_i}{\sum_{i:PD_i \le t} E\,AD_i^{proxy}} \tag{10}$$

$$t^* = \max\{t:\ R_{EL}(t) \le 0.05\} \tag{11}$$

where t is a candidate calibrated-PD approval boundary and t^* is the largest complete approval set satisfying the 5.00% projected principal-loss-to-funded-exposure constraint.

In order to determine the optimal calibration boundary, the strict 5.00% projected principal-loss-to-funded-exposure ratio requirement was used on the 2014 validation cohort. The solver examined all possible PD boundaries, and determined the optimal boundary value to be 0.169611, which resulted in the approval of 110,226 development-period loans with the projected EL rate of 4.8002%.

All PD values calibrated through the isotonic regression include ties, so all loans with the same boundary value were considered equally. Next available PD boundary value was 0.173901, which together with its tie-group of 7,441 loans would have resulted in a portfolio loss rate exceeding the 5.00% constraint with a projected portfolio loss rate of 5.0163%. Therefore, the chosen value of 0.169611 is the maximum feasible complete approval boundary according to the development period rule.

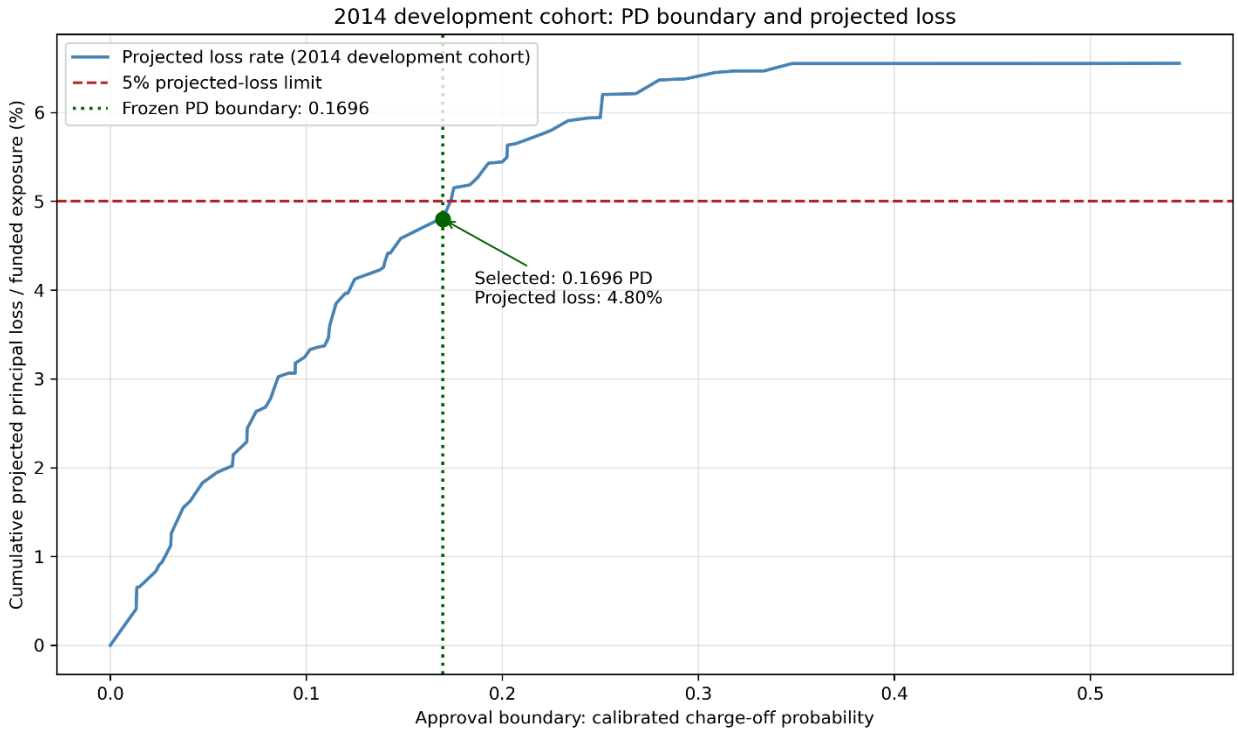


Figure 4.6: 2014 development cohort policy curve. The point of intersection defines the optimal calibrated PD boundary value (0.1696), which allows maximizing the cumulative loan approval volume while meeting the strict 5.00% projected principal-loss constraint.

The 2014 policy-selection results should be seen as the development stage result and not the independent validation result. The isotonic mapping was fit based on 2014 final outcomes, and the 5.00% portfolio constraint was selected based on the same validation cohort. Independence is restored only after applying the selected frozen threshold value to the 2015 OOT cohort.

4.5.1. Out-of-Time Principal-Loss Evaluation

This value of 0.169611 for the PD threshold was then applied to the OOT test cohort from 2015 without any changes. Of 283,026 originated loans from the 2015 OOT test sample, the model has accepted 195,051 loans (with an acceptance ratio of 68.92%), which corresponds to $2.60 billion worth of exposure.

Whereas the expected loss rate for the selected portfolio was 4.75%, the actual loss rate for the portfolio was 5.27% – the gap is 0.52 percentage points. This discrepancy reflects the weakness of using a policy optimized and selected based on development period outcomes when applying it to a subsequent cohort where a different observed risk level prevails. Nevertheless, even this frozen policy still selects a lower-loss portfolio as compared to the accept-all benchmark, whose actual loss rate was 7.36%. This

comparison should be interpreted as a retrospective portfolio-selection result rather than evidence of causal loss prevention.

Metric	Frozen EL Policy	Accept-All
Loans retained	195,051	283,026
Approval/retention rate	68.92%	100.00%
Funded exposure	$2.607 bn	$3.624 bn
Projected principal loss	$123.75 m	—
Projected loss rate	4.75%	—
Realized principal loss	$137.26 m	$266.90 m
Realized principal-loss rate	5.27%	7.36%

Table 4.4: Retrospective 2015 OOT Principal-Loss Evaluation

Thus, the frozen policy has selected a less risky portfolio subset than the historically originated one. However, the gap between the accept-all and the selected portfolios in terms of their actual losses should be treated as a selection criterion, but not necessarily as capital savings, since rejected loans could also bring cash inflows.

4.5.2. Undiscounted Net-Cash Return Analysis

Besides the comparison of principal loss, a cash flow return analysis using undiscounted net cash flows including interest payments, late fees and recoveries was performed. Net cash is calculated as:

$$NetCash_i = TotalPayment_i - CollectionRecoveryFee_i - FundedAmount_i \quad (12)$$

$$NetReturn(S) = \frac{\sum_{i \in S} NetCash_i}{\sum_{i \in S} FundedAmount_i} \quad (13)$$

It is a cash measure that is undiscounted. `Total_pymnt` already includes recoveries, so no separate add-on is included for recoveries. As a result, the return measure doesn't include costs of funding, cost of servicing, taxes, economic capital and the time value of money, which means it cannot be regarded as a measure of economic or accounting profit.

2015 portfolio subset	Loans	Net cash	Net return on funded amount
EL-policy approved	195,051	$187.09 m	**7.18%**
EL-policy rejected	87,975	$43.18 m	**4.24%**
Accept-all	283,026	$230.27 m	**6.35%**

Table 4.5: Undiscounted Net-Cash Outcomes for the Frozen EL Policy and Accept-All Benchmark, 2015 OOT Cohort

Undiscounted net cash returned by the approved 2015 portfolio equaled $187.09 million, that corresponds to 7.18% of return on funded exposure. Return for the simulated rejected subset is 4.24%, so we can conclude that the frozen policy concentrated more funded exposure into the loans with better net cash return. Therefore, the approved portfolio has provided a better return on funded exposure compared to 6.35% of the accept-all portfolio, although in this case it is a result of return efficiency, but not of higher absolute net cash, since the accept-all portfolio contains all originated loans observed.

4.6. Explainability (SHAP Global and Local Explanations)

4.6.1. Global Model Attribution

In order to provide transparency and governance analysis, TreeSHAP (Shapley Additive Explanations) was run on the unweighted LightGBM challenger model (Lundberg & Lee, 2017). The analysis offers reproducible global and local explanations of the model's behavior and does not present itself as enough evidence of regulatory compliance.

Global feature attribution was run using a reproducible sample of 10,000 observations from the 2015 OOT cohort. FICO midpoint, logged annual income, number of credit inquiries, loan amount, DTI and revolving-credit utilisation were some of the largest feature attributions to the fitted model output. The feature attributions describe only the statistical relationships within the fitted LightGBM model; no causal relationship and proof of non-spuriousness is offered (Hadji Misheva et al., 2021). TreeSHAP was run on the unweighted raw LightGBM model. Therefore, the SHAP contribution scores are expressed on the unweighted raw margin/log-odds scale of the LightGBM model and should not be considered as percentage-point changes in the calibrated probability of default.



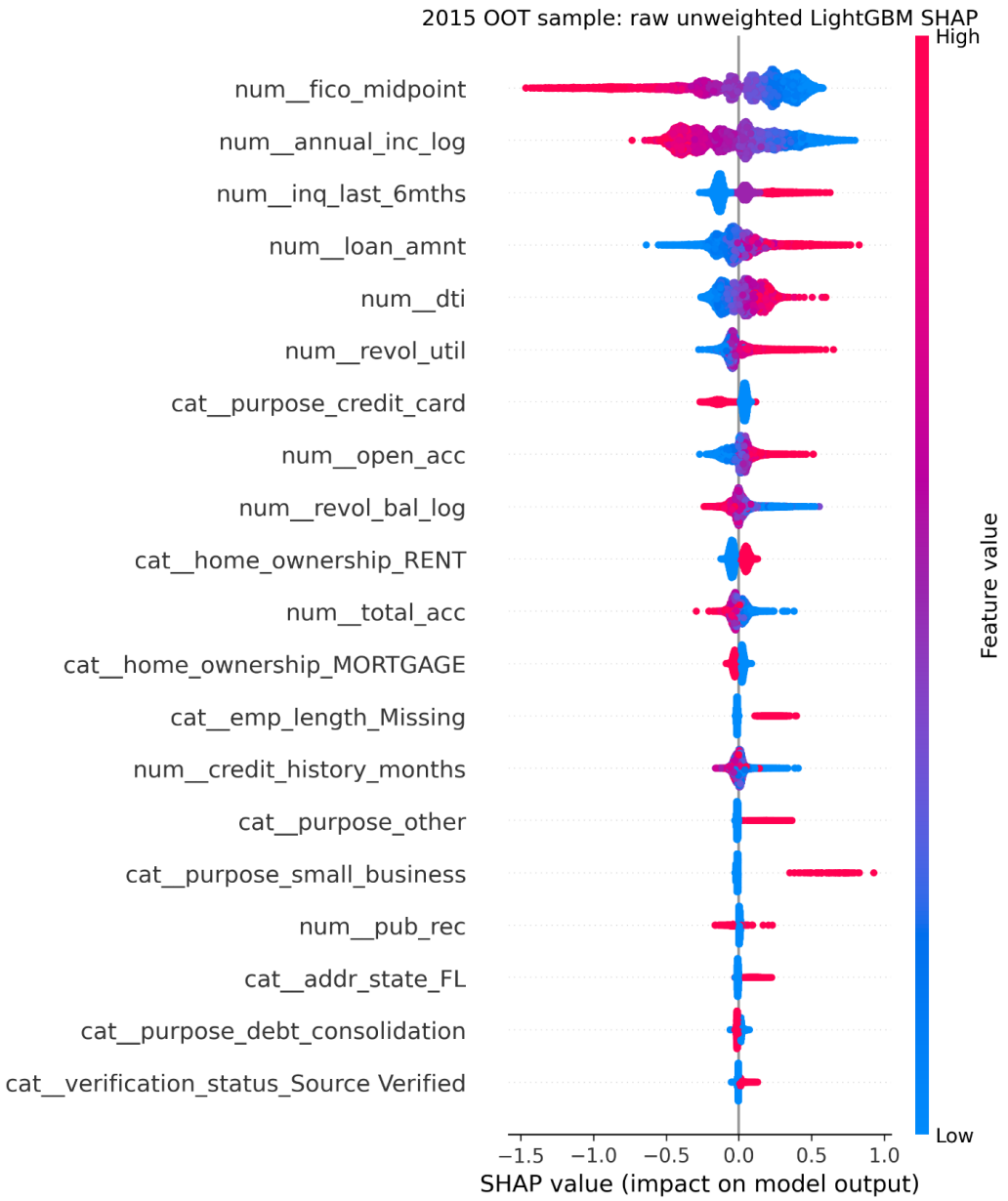


Figure 4.7: Global TreeSHAP feature attributions for the raw unweighted LightGBM model on a reproducible 2015 OOT sample. Feature attributions are sorted by their mean absolute values and show the most important contributors to the model output on the unweighted raw margin/log-odds scale.

4.6.2. Local Technical Explanation

Additionally, SHAP is necessary to provide the local, per-instance explanations that are required for the simulated Adverse Action Notices. Figure 4.8 depicts a reproducibly selected instance that was rejected in 2015. For this example, the underlying LightGBM produces a raw probability of 50.37% that gets calibrated through the frozen isotonic calibration map to a probability of 54.55% of charge-off

occurrence. Both probabilities are above the frozen 16.96% rejection threshold for the post-calibrated probability. The waterfall plot illustrates the individual feature transformations contribution to the raw LightGBM output. It does not break down the isotonic mapping function, nor does it provide the percentage point contributions to the final calibrated probability[6]. Therefore, the explanation should be regarded as a technical model attributions example and not a legal reason for rejection. The adverse-action process in production will need separate validation of reason-codes along with legal and compliance review.

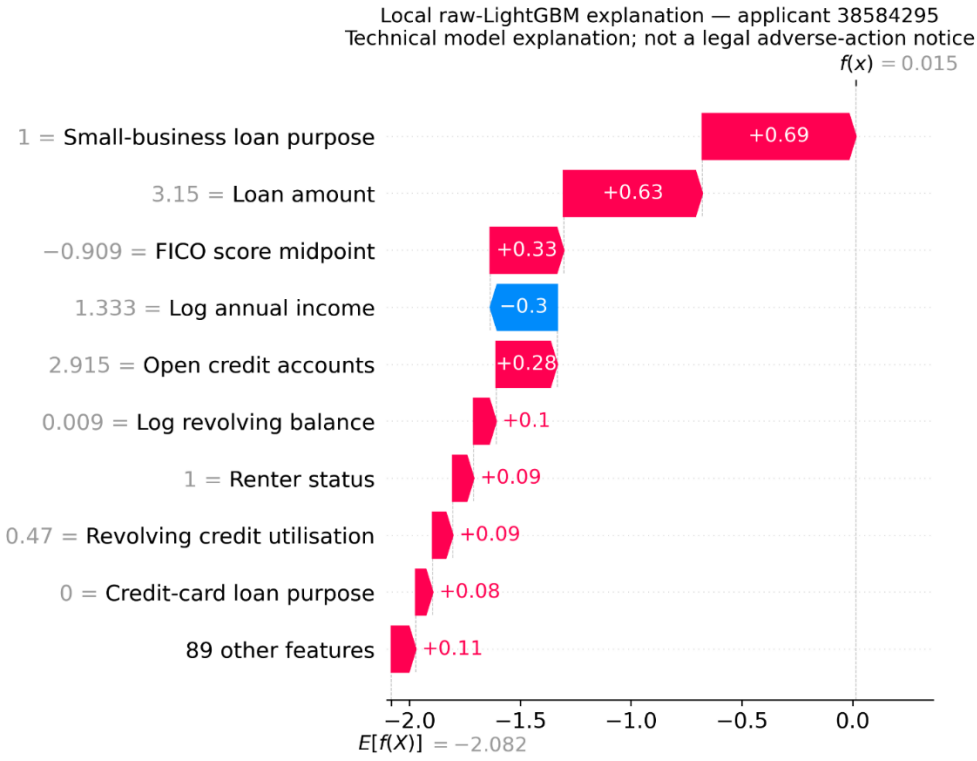


Figure 4.8: TreeSHAP local explanations for a reproducibly chosen rejected 2015 observation. The SHAP contribution values are computed for the unweighted raw margin/log-odds LightGBM output; the final isotonic-calibrated PD and frozen policy threshold are reported separately.

4.7. Proxy-Group Fairness Audit

Also, the frozen 2014 EL policy was assessed to determine any differences in simulated decision outcomes based on the home ownership proxy. The home-ownership proxy is used strictly for illustrative purposes, and the renters will be compared to non-renters. Neither group is used in the protected-class status

[6] TreeSHAP represents the mathematical attribution of the underlying model output. The adverse-action process in production will require the validation of reason-codes separately from legal and compliance reviews. TreeSHAP alone cannot ensure ECOA compliance.

nor in disparate-impact analysis[7]. In accordance with the fairness framework described by Hardt, Price, and Srebro (2016), the differences in simulated decision-error rates between the two proxy groups are considered in this example.

False positive is an historically originated loan that was finally terminated as Fully Paid but would have been rejected according to the frozen EL policy. This measure reflects simulated allocation error within the group of historically originated loans; it does not imply that the borrower was legally "qualified" or denied credit from Lending Club.

Metric	Non-renters	Renters
Loans	161,563	121,463
Observed charge-off rate	12.80%	17.65%
Simulated rejection rate	22.88%	42.00%
FPR	20.38%	38.50%
95% Wilson CI for FPR	[20.17%, 20.59%]	[38.20%, 38.81%]
FNR	60.11%	41.69%
TPR	39.89%	58.31%

Table 4.6: 2015 OOT Proxy-Group Decision Metrics Under the Frozen EL Policy

For renters and non-renters, the FPR ratio is 1.8895. It should also be noted that the common threshold yielded a relatively higher true positive rate (TPR) and a lower false negative rate (FNR) for renters. Therefore, the audit finds the allocation disparity of different error types for two different socioeconomic proxies.

In summary, the results above indicate an allocation disparity within the proxy groups under the frozen policy; however, it does not prove unlawful discrimination or fair-lending non-compliance. Any potential adjustment will be a new policy and must be chosen by using development data and testing on a holdout sample of prospective data in 2015.

[7] Home ownership is used as a proxy for the illustrative socio-economic purpose of this sensitivity audit only and is not considered a legally protected characteristic in any way. The renter/non-renter distinction cannot be considered as protected class or disparate impact analysis.

CHAPTER 5: DISCUSSION

5.1. Retrospective Financial Performance of the Expected-Loss Policy

The EL policy helped to reduce the principal loss rate of the retained 2015 portfolio from the accept-all benchmark of 7.36% to 5.27%, and to increase the undiscounted net return on funded amount from 6.35% to 7.18%. At the same time, the rejected subset itself showed a net positive cash flow of 4.24%, while the accept-all portfolio had more absolute net cash flow. Thus, this finding proves efficient capital allocation under the risk constraint rather than the overall financial advantage.

This finding also demonstrates the distinction between the predictive discrimination and financial decision performance: the predictive ability of a model does not define the consequences of the financial decision based on the model.

5.2. Temporal Drift and Calibration Decay

An important observation of this study is the susceptibility of static probability estimates to longitudinal temporal and portfolio drift. For the Lending Club sample, the observed charge-off rate increased from 12.52% to 14.89% in the period of 2011-2015. Based on the decomposition, it can be stated that the observed increase in the charge-off rate was driven mostly by within-segment deterioration rather than the change in portfolio composition. Since no external macroeconomic factors are used in the decomposition, the exact cause of this deterioration cannot be determined due to macroeconomic conditions.

This resulted in the degradation of the absolute probability calibration of the LightGBM model, although the ranking capability remained generally constant, as the ROC-AUC score stayed relatively steady at roughly 0.60 every month. The 2014-fitted isotonic regression consistently underestimated the observed higher frequency of charge-offs in the 2015 sample. This highlights the necessity for continuous assessment and potential recalibration of the model whenever the relation between its predictions and the observed outcome changes.

5.3. Explainability and Proxy-Group Fairness

TreeSHAP helped to integrate both global feature attributions and instance-level transparency into the model which could be used as technical justification for model governance but did not prove the ECOA compliance (Lundberg & Lee, 2017). However, there were material differences between simulated proxy groups in terms of decision error rates identified via the audit.

For example, under the frozen EL boundary the FPR ratio between the renter and non-renter proxy group was 1.8895. In addition, according to the feature-policy sensitivity experiments, removal of the `home_ownership` predictor reduced the FPR ratio to 1.4977, whereas the realized principal loss rate changed from 5.27% to 5.32%. Thus, the feature policy change had different effect on the predictive, financial and proxy group outcomes. The reason why the result cannot be considered as the evidence of the trade-off is that the renter status is merely socioeconomic proxy and correlated predictors still exist.

5.4. Methodological Limitations and Future Work

Despite the fact that this framework simulates production constraints, there are certain limitations to be noted. First, the model works only with the data of originated loans; the repayment results for rejected customers are unknown, thus causing a selection bias. Second, EAD and LGD were artificially calculated based on historical data on funding amount and recovery rate as a simplified approximation; in the future, the research should concentrate on developing independent predictive ML sub-models for these indicators. Finally, real label maturation delay was approximated, while for the live production model more sophisticated solutions would be required in order to deal with non-resolved loans at the moment of retraining.

Another limitation is connected to the retrospective nature of the financial evaluation. The isotonic regression transformation and the boundary for loan approval rely on the fully resolved results of the 2014 year, which would not have been known at the decision date of January 2015. Additionally, the net-return measure, which is reported above, is not discounted and does not include funding, servicing, capital and timing costs, hence cannot be considered as a measure of profitability. The SHAP values analysis is performed for the raw LightGBM model before isotonic regression transformation, while the fairness audit considers renter status as a socioeconomic variable and not as protected class indicator. Thus, the 5.00% portfolio constraint should be viewed as a development stage risk limit and not as the actual limit on the subsequent losses. As can be seen from the 2015 OOT evaluation, the portfolio which was selected had a predicted loss rate of 4.75%, however, realized principal loss rate was 5.27%. The Expected Loss policy thus is a simulation of decision-making rule rather than proof that the actual losses will be below the defined constraint. Therefore, future work should include point-in-time labeling, predictive LGD/EAD modelling, discounted cash-flow analysis, validated adverse-action reason codes, accepted protected-class or proxy methodology, and prospective fairness-policy evaluation.



CHAPTER 6: CONCLUSION

This paper builds upon the prior static classification framework of Part 1.0 (Arain, 2026) to develop a reproducible and cost-sensitive credit-risk research pipeline based on 603,589 Lending Club loan resolutions and chronological out-of-time (OOT) evaluation. In this work, the empirical evidence confirms a slight discrimination improvement of LightGBM over Logistic Regression that is distinguishable from randomness but is limited by the lack of financial performance contribution of ranking.

By means of calibrated charge-off probabilities combined with the historical assumption of LGD and the funded amount as the measure of exposure, a development-stage Expected Loss policy was designed subject to the constraint of 5.00% expected principal loss. Based on these criteria, the frozen policy selected a boundary of the calibrated PD of 0.1696. Upon application to the 2015 OOT data, this policy selected a portfolio with the realized principal loss of 5.27% compared to 7.36% of the accept-all portfolio and undiscounted return on the funded amount of 7.18% compared to 6.35%. In this way, the results demonstrate improved risk concentration and return efficiency but do not prove maximum economic profitability under the specified policy.

The temporal diagnostics have shown that the stable discrimination performance can go along with the deterioration of probability calibration. TreeSHAP offers a reproducible technical explanation of the underlying LightGBM model, while the renter/non-renter audit helps to identify the material differences in decision error rates under the frozen policy. These results should be interpreted as the model governance diagnostics, not as causal or regulatory conclusions.

In summary, this paper has shown that the useful credit-risk modelling should take into account discrimination, calibration, temporal stability, financial consequences, explainability and sensitivity to fairness criteria. The resulting framework can be viewed as a transparent policy simulation and model-validation research environment rather than a validated production lending system.



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APPENDICES:

Appendix A: Extended Exploratory Data Analysis (EDA) Tables

Feature	Count	Min	1%	50%	99%	Max
loan_amnt	603,589.0	1000.00	1500.00	10000.00	35000.00	35000.00
int_rate	603,589.0	5.32	5.32	11.99	22.15	28.99
annual_inc	603,589.0	0.00	17400.00	60000.00	250000.00	9000000.00
dti	603,587.0	0.00	1.99	17.16	37.15	999.00
fico_range_low	603,589.0	660.00	660.00	685.00	795.00	845.00
fico_range_high	603,589.0	664.00	664.00	689.00	799.00	850.00
delinq_2yrs	603,589.0	0.00	0.00	0.00	4.00	30.00
inq_last_6mths	603,589.0	0.00	0.00	0.00	4.00	8.00
open_acc	603,589.0	0.00	3.00	10.00	28.00	84.00
pub_rec	603,589.0	0.00	0.00	0.00	2.00	86.00
revol_bal	603,589.0	0.00	339.00	10685.00	89175.48	2904836.00
revol_util	603,282.0	0.00	2.40	54.50	98.20	892.30
total_acc	603,589.0	2.00	5.00	23.00	59.00	169.00

Table A.1: Numerical Feature Summary (2011–2015 Cohort) *This table describes the numerical feature distribution among 603,589 resolved loans, using extreme percentiles used to detect right-skewness and outliers before taking the logarithm.*

Category Group	Value	Total Loans	Charged Off	Charge-Off Rate (%)
Home Ownership	RENT	260,580	42,955	16.48%
	OWN	61,632	9,186	14.90%
	MORTGAGE	281,300	32,073	11.40%
Verification Status	Verified	181,498	29,527	16.27%
	Source Verified	215,551	30,709	14.25%
	Not Verified	206,540	23,991	11.62%
Employment Length	NaN (Missing)	35,885	7,351	20.48%
	< 1 year	49,273	7,166	14.54%
	1 to 9 years	330,831	46,146	~13.90%
	10+ years	187,601	23,567	12.56%
Loan Purpose (Top 5)	debt_consolidation	344,289	50,371	14.63%
	credit_card	148,087	17,260	11.66%
	home_improvement	34,606	4,296	12.41%
	other	31,838	5,194	16.31%
	major_purchase	11,986	1,520	12.68%

Table A.2: Categorical Feature Distributions and Base Charge-Off Rates *This table illustrates the size of the population and charge-off probability for various categorical features in the whole dataset.*

Appendix B: Complete Baseline Feature Schema

In order to ensure data leakage doesn't occur in the pipeline, and models rely only in the information available at the time of origination, more than 120 columns were stripped away systematically from the original 151-feature Lending Club dataset.

Excluded Variables (Examples):

Exclusion Category	Variables	Reason for Exclusion
Target Leakage / Post-Origination Fields	`funded_amnt`, `total_rec_prncp`, `recoveries`, `collection_recovery_fee`, `total_pymnt`	Excluded to prevent target leakage and avoid using information unavailable at the time of loan origination.
Lender Pricing Outputs	`grade`, `sub_grade`, `int_rate`, `installment`	Excluded to avoid reliance on lender-assigned risk assessments and pricing decisions.
Administrative Identifiers	`id`, `issue_d`, `loan_status`	Excluded as identifiers, administrative fields, or outcome-related variables rather than independent predictive inputs.

Table B.1: Excluded Variables and Rationale

Frozen Baseline Feature Policy (21 Inputs):

The last step in the model utilized 21 engineered input features.

No.	Feature Name	Description / Transformation
1	`loan_amnt`	Original loan amount requested
2	`annual_inc_log`	Log-transformed `annual_inc`
3	`dti`	Debt-to-income ratio
4	`delinq_2yrs`	Number of delinquencies in the past two years
5	`inq_last_6mths`	Number of credit inquiries in the past six months
6	`open_acc`	Number of open credit accounts
7	`pub_rec`	Number of public derogatory records
8	`revol_bal_log`	Log-transformed `revol_bal`
9	`revol_util`	Revolving credit utilization rate
10	`total_acc`	Total number of credit accounts
11	`collections_12_mths_ex_med`	Number of collections in the past 12 months, excluding medical collections
12	`acc_now_delinq`	Number of accounts currently delinquent
13	`pub_rec_bankruptcies`	Number of public bankruptcy records
14	`credit_history_months`	Derived mechanically from `earliest_cr_line` and `issue_month`
15	`fico_midpoint`	Calculated as the average of `fico_range_low` and `fico_range_high`

Table B.2: Frozen Baseline Feature Policy: Numerical Features (15)

No.	Feature Name	Description / Treatment
1	`emp_length`	Employment length; missing values retained as an explicit "Missing" category
2	`home_ownership`	Home ownership status
3	`verification_status`	Income verification status
4	`purpose`	Stated purpose of the loan
5	`application_type`	Type of loan application
6	`addr_state`	Borrower's state of residence

Table B.3: Frozen Baseline Feature Policy: Categorical Features (6)

The frozen baseline feature set consists of 21 engineered input features, consisting of 15 numerical and 6 categorical features. Variables were excluded to limit any target leakage and the use of lender-derived output features, thereby ensuring the predictor only makes use of relevant borrower and loan attributes.



Appendix C: Model Hyperparameters and Environment

For consistency in replicating the results in an academic context during training, validating, and testing, the models were trained with consistent random seeds and hyperparameters. No exhaustive hyperparameter grid-search optimization took place, showing that the strong performance of the non-linear challenger is due to the architecture and the chronological design rather than extensive over-tuning.

Model	Implementation	Key Hyperparameters

LightGBM Challenger	`lightgbm.LGBMClassifier`	`n_estimators = 200`
		`learning_rate = 0.05`
		`random_state = 42`
		`n_jobs = -1`
		`verbosity = -1`

Logistic Regression Baseline	`sklearn.linear_model.LogisticRegression`	`max_iter = 1000` `random_state = 42`

Probability Calibrator	`sklearn.calibration.CalibratedClassifierCV`	`method = "isotonic"`
		`cv = "prefit"`
		(Fitted strictly on 2014 Validation set)

Table C.1: Fixed Model Hyperparameters

DYNAMIC EXPECTED-LOSS Optimization in Credit Risk

A Chronological, Explainable, and Proxy-Fair ML Framework

THESIS 2.0 AT A GLANCE

QUESTION

- How can credit-risk ML move beyond static classification into a calibrated Expected Loss decision framework?
- How should drift, financial consequences, explainability, and proxy-group fairness be evaluated together?

DATA

- Lending Club (Resolved 36m Loans) with 603,589 resolved loans.
- Target: final-status charge-off outcome.
- 21 baseline pre-origination features; repayment outcomes isolated for retrospective evaluation.

DESIGN

- Chronological split: Train 2011–2013, Validate & Recalibrate 2014, Frozen OOT Test 2015.
- Historical policy simulation rather than a live production deployment.

METHODS PIPELINE

1. Chronological Split
2. Fit Raw LightGBM
3. Isotonic Calibration
4. Historical LGD + EAD Proxy
5. Optimize EL Boundary
6. Audit Drift, SHAP & Fairness

RESULTS SUMMARY

MODEL PERFORMANCE

- LightGBM achieved OOT ROC-AUC = 0.6625, compared with 0.6569 for Logistic Regression.
- The performance gain was modest but consistent.
- Probability calibration required separate assessment beyond ranking performance.

EXPECTED LOSS POLICY

- A 5.00% projected principal-loss constraint selected the calibrated-PD boundary of 0.1696.
- Applied unchanged to 2015, the frozen policy retained 68.92% of originated loans.
- The retained portfolio realized 5.27% principal loss and 7.18% return on funded amount.

AUC COMPARISON

0.6569 Logistic Regression

0.6625 LightGBM

PORTFOLIO COMPARISON

6.35% Accept-All Return

7.18% EL Policy Return

0.1696 PD Approval Threshold

5.00% Projected EL Limit

5.27% Realized Loss Rate

1.8895 Renter / Non-renter FPR Ratio

DRIFT & EXPLAINABILITY

- Kitagawa decomposition indicated that the observed charge-off increase was dominated by within-segment deterioration rather than portfolio-mix change.
- TreeSHAP identified key contributors to the raw LightGBM output, including FICO, annual income, DTI, and related risk variables.

PROXY-GROUP FAIRNESS AUDIT

- Under the frozen policy, renters had a higher False Positive Rate than non-renters (38.5% vs 20.4%).
- The resulting FPR ratio was 1.8895.
- Renter status was used only as an illustrative socioeconomic proxy, not as a protected-class attribute.

TAKEAWAY

- Expected Loss optimization improved retrospective portfolio selection relative to an accept-all benchmark.
- Stable discrimination did not guarantee stable calibration over time.
- Credit-risk ML should be evaluated jointly on discrimination, calibration, temporal stability, financial consequences, explainability, and proxy-group fairness.

LIMITATIONS

- Originated loans only (selection bias).
- Historical LGD and funded amount used as simplified proxies.
- Home ownership is an illustrative socioeconomic proxy.
- Retrospective design with delayed-label assumptions.

Reproducibility: Python 3.13.5 | scikit-learn 1.6.1 | LightGBM | shap 0.50.0 | seed = 42

Sana Ur Rehman Arain | 2026